

# BIOTECHNOLOGY: BASIC LAB TECHNIQUES



*Victor Pham*  
Irvine Valley College

# Biotechnology: Basic Lab Techniques

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Detailed Licensing

## Licensing

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*A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).*

# 1: Introduction to Biotechnology

## How To Use This Lab Manual:

This lab manual should be used along with the Textbook: Biotechnology Techniques: From Theory to Practice BIOT173A.

## Learning Objectives

By the end of this lab, you should be able to:

- Define biotechnology and describe its interdisciplinary nature.
- Identify key applications of biotechnology in medicine, agriculture, environmental conservation, and forensics.
- Recognize ethical concerns associated with biotechnological advancements.

## Definition: Term

Review and familiarize yourself with the following key terms:

- Biotechnology
- Genetic Engineering
- Genetically Modified Organisms (GMOs)
- Monoclonal Antibodies
- Biofuels
- DNA Analysis
- Personalized Medicine
- Bioremediation
- Ethics in Biotechnology

## ? Guiding Questions:

Answer the following questions in your lab notebook before class:

- How do biotechnology and genetic engineering differ?
- What are three key areas where biotechnology is making a major impact?
- Why is genetic modification both beneficial and controversial?
- What role does biotechnology play in addressing environmental issues?
- What are some ethical concerns associated with genetic data and privacy?

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## Chapter One

### Introduction to Biotechnology

Welcome to the fascinating world of biotechnology! In this incredible journey, we'll explore how the perfect blend of biology and technology takes scientific knowledge and turns it into practical solutions, all while keeping a close eye on ethics. So, what is biotechnology all about? Think of it as a marriage between biology and technology. We delve into the amazing potential of living organisms, investigating their cellular and molecular processes to create innovative solutions for real-world challenges. It's not just about scientific exploration; it's a collaborative effort to use our understanding of biological systems to develop groundbreaking applications that impact various aspects of our lives. Now, let's break it down. **Biotechnology** involves understanding living things at the cellular and molecular level and using that knowledge to solve problems. And guess what? I've got a cool figure (Figure 1.1) that'll give you a quick glimpse of the wide-ranging scope in biotech.

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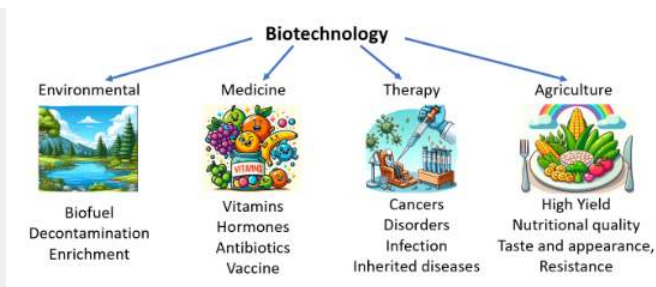


Figure 1.1: A Comprehensive Figure Exploring Environmental, Therapeutic, Medicine, and Agricultural Applications. This part of the figure shows how biotechnology is making a big impact on the **environment**. Check out biofuel production – it's using genetically modified microorganisms to create sustainable fuel options, reducing our reliance on fossil fuels. Ever wondered how we clean up polluted soil and water? Genetically engineered bacteria are breaking down pollutants, showing how biotechnology plays a crucial role in environmental cleanup. Now, let's zoom in on **healthcare therapy**. Ever heard of monoclonal antibodies like Rituximab? They're fighting cancer with precision and effectiveness. For disorders like diabetes, we're can genetically modified cells to produce essential proteins like insulin, changing how we treat these conditions. Moving on to **medicine** – this part of the figure dives into the production of high-vitamins, hormones, antibiotics, and vaccines. Biotechnological processes are at play, making food with boosted nutritional value and essential compounds like insulin. Antibiotics are getting a makeover through genetic engineering, and vaccines are being created using biotechnological methods, playing a crucial role in preventing diseases. Last but not least, let's explore the future of **farming**. See those high-yield crops? Genetically modified organisms are boosting productivity. Nutrition quality is in the spotlight with crops featuring improved nutritional content. Taste and appearance? Biotechnological interventions are enhancing the sensory aspects of crops. And don't forget about crop resistance to pests – ensuring sustainable agriculture.

In this Chapter, we'll dive into how biotechnology is transforming the medical landscape, impacting agriculture, contributing to environmental conservation, aiding forensics, and raising some thought-provoking ethical questions. Let's start in the medical field. Biotechnology is changing the game in treatments. Take insulin, for example – a lifeline for those with diabetes. Guess what? It's produced through genetic engineering. And that's just the tip of the iceberg. Biotechnology is a key player in personalized medicine, tailoring treatments to fit individual genetic profiles. Now, shift your focus to agriculture. Genetically modified organisms (GMOs) are engineered to have specific traits that boost crop yields, improve nutritional content, and resist pests. Check out the banana – once vulnerable to fungi, now standing strong thanks to biotechnological intervention. Biotechnology is also a hero in environmental conservation. Imagine genetically modified bacteria coming to the rescue during oil spills, and biofuels providing sustainable alternatives to fossil fuels. It's all about using science to make a positive impact on our planet. In the world of forensics, biotechnology is a powerful ally in solving crimes. DNA analysis, a cutting-edge tool, helps unravel genetic puzzles during criminal investigations, bringing us closer to the truth. But wait, there's more to the story. Biotechnology isn't just about innovation; it sparks important ethical questions. Think about companies like 23&me – they deal with genetic data, raising debates about privacy and the potential misuse of sensitive information. And when it comes to genetic modifications, especially without proper authorization, scrutiny and ethical concerns come into play.

With great power comes great responsibility. Biotechnology! It's like a team of superheroes – researchers, scientists, and industry experts – coming together from different fields, pooling their knowledge to push the boundaries of what we can achieve. This collaborative and interdisciplinary field is all about the fusion of biology and technology, creating solutions that tackle real-world challenges head-on.

In a nutshell, biotechnology is a dynamic and transformative force, making waves in medicine, agriculture, environmental conservation, and forensics. The examples of insulin, GMOs, and environmental cleanups showcase its incredible power. However, as we navigate this exciting field, let's do so with care and foresight, acknowledging the responsibilities that come with such transformative advancements.

### ? Post-Lab Questions

After completing the lab, reflect on the following questions:

- What is one new concept you learned about biotechnology that you found surprising or particularly interesting?
- How does biotechnology influence both scientific progress and societal issues?
- What ethical considerations should scientists keep in mind when developing new biotechnological applications?
- In your own words, summarize the role of biotechnology in shaping the future of medicine, agriculture, and environmental sustainability.

### 📌 Exit Ticket

- One key takeaway from today's discussion.
- One question you still have about biotechnology.
- One potential career or industry where biotechnology is applied that interests you.

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## 2: Scientific Method

### Pre-Lab: The Scientific Method

#### Learning Objectives

By the end of this lab, students should be able to:

- Define and explain the scientific method.
- Differentiate between testable and non-testable hypotheses.
- Understand the importance of falsifiability in science.
- Recognize biases and emotional responses in scientific reasoning.

#### Background Information:

Science is built on systematic observation, experimentation, and reasoning. The scientific method provides a structured way to investigate the world by forming hypotheses, conducting experiments, and analyzing results. A key principle in science is falsifiability—the ability to prove an idea wrong through evidence. Without falsifiability, an idea falls outside the realm of science.

#### Key Concepts to Review Before Lab:

- **Scientific Method Steps:** Observation, Question, Hypothesis, Experiment, Data Analysis, Conclusion.
- **Falsifiability:** Why testable ideas are fundamental in scientific research.
- **Cognitive Biases:** How emotional responses can mislead scientific reasoning.
- **Examples of Scientific Reasoning:** Red Dragon Hypothesis, Dihydrogen Monoxide example.

#### Pre-Lab Questions:

- Why is falsifiability important in science?
- How does the Red Dragon Hypothesis illustrate the concept of falsifiability?
- Describe an example where emotional bias might influence scientific conclusions.
- Formulate a testable hypothesis for the question: "Does listening to music while studying improve test performance?"

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## Chapter Two

### The Scientific Methods

Alright, let's dive into the world of science! To grasp the concept, we'll begin with the main idea – **science** is a systematic approach to understanding the natural world by observing and experimenting. It's like following steps to cook a food dish, but for scientists, these steps ensure their findings are **reliable** and **accurate**. Now, at the heart of it all is the scientific method. This method tells scientists that the ideas they come up with, called **hypotheses**, need to be both **testable** and **falsifiable**. What does that mean? It means the conclusions they draw should be open to scrutiny and could be proven wrong. This sets science apart from beliefs that can't be tested. It's like having a recipe that can be tasted and adjusted to make sure it's just right. So, as we explore science, remember, it's all about observing, experimenting, and making sure our ideas can stand up to testing and scrutiny.

Why is **falsifiability** important in science? Imagine the **Red Dragon Hypothesis** (By Carl Sagan) – someone claims to have an invisible red dragon in their garage. Now, no matter what questions you throw at them, they can come up with explanations to justify the dragon's existence. This scenario highlights the difficulty of conducting scientific investigations on ideas that can't be proven wrong. Now, this is where the importance of **falsifiability** comes in. It emphasizes why hypotheses must be testable and open to the possibility of being proven incorrect. Think of it like this – in science, we want our ideas to be like recipes that can be tasted and adjusted. We need to be able to test and scrutinize them to ensure they stand up to examination. By insisting on falsifiability, science stays grounded in what can be observed and tested. This sets it apart from ideas that can't be checked, helping us maintain a solid foundation in the pursuit of knowledge. So, always remember, in the world of science, we want ideas that can be tested and potentially proven wrong – that's what keeps our investigations reliable and grounded.



The Lion Image by ChatGPT. The Window Image by <https://unsplash.com/@matheusfrade>. The Car Image by <https://unsplash.com/@aahubs>

**Figure: The Pervasive Phenomenon.** This figure explores our innate tendency to perceive meaningful patterns, particularly faces, in random stimuli. The image depicts a scenario where a lion is camouflaged in the grass, triggering our instinct to recognize faces for survival. However, what if there's just a shadow resembling a face and no actual lion? Look closely, and you might notice faces in unexpected places – a car's grille, the pattern on a door, or even the arrangement of clouds in the sky. This illustrates how our brains are wired to seek familiar patterns, sometimes leading us to see faces where none actually exist.

Before we explore the details of science, let's delve into why studying science is so crucial. If we rewind the clock 60,000 years to our ancestors, survival was all about recognizing potential dangers. Even today, those survival instincts impact how we perceive things, sometimes leading us to believe in ideas that aren't quite true. Our tendency to see patterns and meanings where there are none can result in misguided beliefs – think horoscopes, illusions, or mythical creatures. To illustrate this, let's consider an example in **figure 2.1**. Remember that picture with the camouflage of a lion in the grass? Most students say they see a face first, and when asked if they would run away, many respond with a yes. Now, here's the twist – what if there's just a shadow resembling a face, and no actual lion? Would they still run away? The people back then who chose to run, even if there wasn't a lion, represent our ancestors who succeeded in seeing faces, a trait likely linked to their survival. This natural bias to see faces on objects, even when they don't exist, is deeply ingrained in us.

**Dihydrogen monoxide** is found in **all** ocean **toxin** and **pollutants**.

High dose of **dihydrogen monoxide** is **extremely toxic** to the human body



**Should we invest to purify dihydrogen monoxide in our drinks?**

The Climbing from Ocean Image by <https://unsplash.com/@dallehj>

**Figure: Beware of Dihydrogen Monoxide!** It's in our homes, our bodies, everywhere! This colorless, odorless liquid is linked to 100% of drowning cases and Hyponatremia. Stay vigilant – ban Dihydrogen Monoxide!

Alright, let's dive into the **scientific method**, which is more than just a set of steps – it's a way of thinking. Encouraging critical thinking involves questioning information that comes our way. To illustrate this, we'll explore the **Dihydrogen Monoxide scenario** in **figure 2.2**. In a classroom setting, we used the example involving dihydrogen monoxide, which is essentially water. This example took the form of an exaggerated satire commercial, portraying dihydrogen monoxide as hazardous and toxic, similar to

carbon monoxide. The use of alarming terms aimed to trigger an emotional response. Following this, students were asked if they would like to purify our drinks from dihydrogen monoxide. Surprisingly, most students answered yes. However, this immediate reaction was driven by an **emotional bias**. They skipped crucial steps in the scientific method, jumping straight to a conclusion. This scenario underscores the importance of starting with the fundamental step of **asking questions** before drawing conclusions. By dissecting emotional manipulation in seemingly factual claims, we equip ourselves to navigate information more critically. As we navigate the scientific method, remember the significance of questioning and not letting emotions cloud our judgment

The initiation of the scientific journey hinges on asking the right questions. Not every question fits the scientific mold – the key is to formulate questions that are both **testable** and **falsifiable**. For example, asking whether a student is possessed by a demon might be intriguing, but it lacks the attributes needed for scientific inquiry. The concept of a demon is not well-defined or measurable, making it impossible to test or falsify. On the flip side, a question like "How tall is my trashcan?" is an excellent example. It's specific, measurable, and can be tested and falsified based on concrete observations.

Now, onto the next crucial step in the scientific method – formulating a **hypothesis**. While some of you might be familiar with the "if" and "then" structure, it essentially boils down to making an educated guess derived from observations. For instance, if we're exploring the height of a trashcan, a hypothesis could be as straightforward as "I think the trashcan is about 3 feet tall based on my observation." This sets the stage for further investigation and represents our initial educated guess. Testing the height of the trashcan can be done using tools like a ruler or measuring tape, providing tangible data that either supports or challenges the initial hypothesis. This systematic approach ensures that scientific inquiry is grounded in empirical evidence, fostering a disciplined and reliable way for us to understand the world around us.

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### Summary

During this lab, students applied the scientific method to assess testable hypotheses. They also examined cognitive biases and the importance of falsifiability. Key examples, such as the Red Dragon Hypothesis and the Dihydrogen Monoxide scenario, were discussed to reinforce critical thinking.

The scientific method is more than a classroom exercise—it is a fundamental approach to understanding reality. Whether evaluating medical treatments, analyzing news reports, or making everyday decisions, applying the principles of testability and falsifiability ensures we rely on evidence rather than assumptions. By critically analyzing information and questioning biases, we can make informed choices in both science and life.

### Post-Lab Questions

1. Reflect on an instance in today's lab where you or a classmate encountered bias. How did it affect the scientific process?
2. Why is it important to ask testable and falsifiable questions in science?
3. If you had to explain the scientific method to someone unfamiliar with science, how would you describe it?
4. What improvements would you suggest to make scientific inquiry more rigorous in everyday decision-making?

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### 3: Lab Safety

#### Learning Objectives

- To understand the fundamental principles of laboratory safety, including hazard identification, proper waste disposal, personal protective equipment (PPE), and emergency procedures.
- To identify and locate safety equipment within the laboratory setting.

#### Background Information:

Laboratory safety is essential for preventing accidents and ensuring a secure environment for scientific experimentation. Universal hazard labels, PPE, and proper waste disposal methods are critical components of laboratory safety. Additionally, understanding emergency protocols and maintaining personal hygiene further contribute to a safe workspace.

#### Pre-Lab Questions:

1. What is the purpose of Personal Protective Equipment (PPE) in the laboratory? List at least three examples.
2. How do universal hazard labels help in maintaining safety in the laboratory? Describe the significance of each color and number in the hazard rating system.
3. Explain the importance of proper waste disposal. What are the different categories of waste in the laboratory?
4. Why is it essential to know the location of emergency safety equipment, such as fire extinguishers and eyewash stations?
5. What are some personal hygiene practices that should be followed in the laboratory?

- Review Figures 3.1 and 3.2 on universal hazard labels and safety signs.
- Identify the PPE required for laboratory activities and ensure you have the appropriate attire before entering the lab.
- Walk around the lab space to familiarize yourself with the locations of fire extinguishers, emergency showers, eyewash stations, and first aid kits.

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## Chapter Three

### Lab Safety

Alright, let's delve into the essential topic of **laboratory safety**. Its primary aim is to ensure the well-being of individuals and maintain the integrity of experiments. Now, in fields like microbiology, biochemistry, and biotechnology, where inherent hazards are present, implementing strong safety measures becomes absolutely crucial. In this chapter, we'll explore key principles of lab safety, placing emphasis on universal hazard labels, the correct disposal of waste, maintaining personal hygiene, and considering broader safety aspects in the laboratory setting. These principles are foundational to conducting experiments safely and responsibly in the world of science.

Let's talk about a crucial aspect of lab safety – **Personal Protective Equipment**, or PPE. Just like a lifeguard wears a bathing suit and uses a floating device as PPE, in the laboratory, we have specific attire to ensure our safety. For scientific experiments, the appropriate PPE includes closed-toe shoes, long pants, and ensuring that arms are not covered. Think of it as our scientific uniform for protection. Closed-toe shoes provide a barrier against spills or accidents, and long pants protect our legs from potential hazards. It's all about minimizing exposure and ensuring a safe environment. Now, let's draw a parallel to everyday situations. When you go dancing or attend a prom, your PPE is a stylish dress or a sharp suit. Just like the lab requires specific attire for safety, different situations call for different types of protective gear. So, remember, in the laboratory, our PPE includes closed-toe shoes, long pants, and uncovered arms to keep us safe as we delve into scientific experiments.



Figure 3.1: Universal Hazard Labels This figure illustrates universal hazard labels in the laboratory, distinguished by colors and a numerical scale from 0 to 4. Blue indicates health hazards, red signifies flammability, and yellow denotes reactivity. The numerical scale quantifies the severity of each hazard, with "4" indicating high risk and "0" suggesting minimal danger. These labels provide a straightforward and universal means for researchers to assess potential hazards associated with substances, fostering awareness and precaution in the laboratory.

Alright, let's delve the use of **universal hazard labels** (Figure 3.1), similar to traffic signs on the road. These labels, marked by colors and numerical indicators, act as visual guides for researchers, creating a safer working environment. Now, these labels are not just arbitrary; they convey specific information about potential dangers associated with substances in the laboratory. Imagine them as traffic signs in the lab – blue signifies health hazards, red denotes flammability, and yellow indicates reactivity. For instance, a **blue label** on a chemical like benzene warns of health hazards affecting the respiratory system, while a **red label** on acetone indicates its flammability, requiring careful handling away from open flames. Similarly, sodium metal, marked with a **yellow label**, signifies high reactivity, especially with water, leading to a potentially violent reaction. These safety signs are like a universal language in the lab, transcending barriers and promoting a culture of safety. Whether indicating emergency exits or cautioning against specific actions, these signs, like traffic signs on the road, are simple, universal visual cues that convey essential information, minimizing the risk of accidents in the laboratory. Other safety signs can be seen in figure 3.2.



Figure 1: Safety Signs in the Laboratory. In this figure, we're exploring various safety signs commonly found in the laboratory. Each sign comes with a distinctive symbol, representing specific warnings such as environment, toxic, exclamation, flammable, corrosive, and explosive. These symbols are designed to be visual cues, already implying a sense of danger. They're like a universal language in the lab, transcending language barriers and effectively conveying crucial safety information. As teachers and students, recognizing and following these signs is essential for creating a safe and secure laboratory environment. The implied danger in these symbols serves as a reminder to exercise caution and adhere to established safety protocols during experiments.

Alright, let's delve into the importance of responsible disposal of laboratory waste for maintaining a safe working environment. We have different categories of waste, and each requires specific handling. Firstly, we have **Biohazard Waste**: This includes used cultures of infectious bacteria. It's crucial to discard them in designated bins for proper disposal, preventing the potential spread of harmful microorganisms. These materials undergo autoclaving, which is like a pressure cooker, ensuring thorough sterilization. Next up is **Sharp Waste**: This involves broken glassware or used needles. To ensure safe collection and recycling, we dispose of them in specialized containers, reducing the risk of injuries. Now, let's talk about **Hazardous Chemical Waste**: Take hydrochloric

acid, for example. It's considered hazardous and needs careful neutralization before disposal to prevent environmental contamination.

Switching gears, **personal hygiene** in the lab is essential. We need to refrain from eating or drinking in the workspace. The lab environment may contain pathogens that could contaminate our food or beverages, posing health risks. By maintaining cleanliness and being mindful, we transform the laboratory into a safer and more hygienic space. Additionally, keeping the **workspace neat and organized** is more than just about aesthetics; it directly contributes to a safer environment. When personal belongings are left unattended, they can become trip hazards. Picture this scenario: someone walking by might trip over a bag left in the middle of the floor. To avoid such incidents, it's crucial to tuck away or properly store items when they're not in use.

Alright, let's dive into the broader aspects of laboratory safety that involve everyone from the laboratory worker to the owner, institution, and regulatory agencies. Safety is a collective responsibility that extends to all individuals within the laboratory environment.

Now, understanding basic safety rules, operating equipment, and having knowledge of emergency procedures are crucial for everyone in the lab. We maintain controlled access to ensure safety, meaning no unauthorized guests are allowed. Being aware of emergency procedures, such as evacuation protocols, is fundamental for the well-being of everyone in the laboratory.

Personal hygiene is also a key factor. We refrain from activities like smoking, eating, drinking, chewing gum, or applying cosmetics in the lab. Wearing appropriate protective clothing, including long pants, closed-toe shoes, lab coats, and goggles, helps mitigate potential risks. Proper disposal of gloves and avoiding wearing lab coats and gloves in public areas prevent cross-contamination.

Taking a proactive safety approach involves reporting all accidents to a laboratory supervisor. The Hazard Communication Standard (HCS) labels on chemical containers provide a quick assessment of hazards, ensuring a standardized understanding of chemical risks.

When it comes to waste disposal, it's integral to follow guidelines at various levels. Understanding hazardous chemicals and proper storage is crucial. The NFPA Hazard Rating System and Safety Data Sheets provide essential information about chemical hazards and safe handling practices.

- **Guest Policy:** Prohibiting unauthorized individuals from entering the lab ensures that only trained and knowledgeable personnel are present, minimizing the risk of accidents.
- **Emergency Procedures:** Specific emergency procedures, like designated meeting points during fire evacuations, ensure a clear and organized response.
- **Hazard Communication Standards:** Labels on chemical containers provide specific information, ensuring a consistent understanding among researchers.
- **Waste Disposal Guidelines:** Following specific guidelines for hazardous waste disposal ensures compliance with regulations, preventing environmental contamination.
- **NFPA Hazard Rating System:** Identifying chemicals with high flammability through the NFPA diamond-shaped label alerts emergency responders to the severity of the hazard.
- **Safety Data Sheets (SDS):** For example, the SDS for sulfuric acid provides detailed information on properties, hazards, safe handling practices, and emergency control measures.

Remember, laboratory safety requires collaborative efforts, and instilling these principles in the next generation ensures a safe and productive scientific future.

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## LAB ACTIVITY - PART 1 - Safety Inspection Location

Knowing the location of safety equipment in a laboratory is crucial for maintaining a safe working environment. Here are some key reasons why it's important to be aware of the whereabouts of safety equipment:

- **Emergency Response:** This includes items such as fire extinguishers, eyewash stations, emergency showers, and first aid kits.
- **Minimizing Risks:** Being aware of the location of safety equipment helps in minimizing risks by allowing immediate access to tools that can mitigate the impact of accidents.
- **Time Efficiency:** Knowing the location of safety equipment allows for a swift response, potentially reducing the severity of injuries or damage. Every second counts.



- **Maintaining Equipment Readiness:** Regular checks and maintenance of safety equipment are essential, ensuring that everything is in working order when needed.

Please complete the pictures below by indicating the locations of the following safety materials in your laboratory. This exercise is designed to reinforce your knowledge of safety equipment placement, promoting a safer work environment.

Conducted by: \_\_\_\_\_

Date of inspection: \_\_\_\_\_

Campus and Room Location: \_\_\_\_\_

| Safety Equipment                                | Location |
|-------------------------------------------------|----------|
| Eye Wash                                        |          |
| Safety Shower                                   |          |
| Broken Glass waste container                    |          |
| Lab Coat                                        |          |
| Goggles                                         |          |
| Gloves                                          |          |
| Disinfect (Bleach, Ethanol, or Bactericide)     |          |
| Fire Extinguisher                               |          |
| First Aid Kit                                   |          |
| Broom and Dustpan                               |          |
| Fire Alarms                                     |          |
| Exits                                           |          |
| Campus Emergency Number or Campus Police Number |          |

Sketch an image of the lab, and indicate the location of the listed items. Then cut out and paste the image onto your laboratory notebook with appropriate title and labels.

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### ? Reflection Questions:

- Describe one new laboratory safety rule you learned during this lab. How does it help ensure safety?
- Were you able to identify all the safety equipment in the lab? Which piece of equipment was the hardest to find, and why?
- If an accident occurs in the lab, what steps should be taken to report and address it?
- How does understanding hazard labels impact your approach to handling chemicals?
- Reflect on the importance of proper waste disposal. What might be the consequences of improper waste disposal in a laboratory setting?

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## 4: Lab Notebook

### Learning Objectives

- Understand the critical role of lab notebooks in scientific documentation, intellectual property protection, and regulatory compliance.
- Learn the key components of a well-structured lab notebook, including SOPs, QA, and QC.
- Recognize the importance of proper donning and doffing procedures to ensure safety and maintain laboratory standards.
- Understanding lab notebook structure and documentation best practices.
- Writing a Standard Operating Procedure (SOP) for donning and doffing PPE.
- Practicing safety measures to maintain sterile and contamination-free laboratory conditions.

### Background

Lab notebooks are essential tools in research and industry settings, serving as legal documents that establish ownership of intellectual property and ensure consistency in scientific procedures. In biotechnology and other scientific fields, following Standard Operating Procedures (SOPs) guarantees uniformity in data collection. Additionally, implementing Quality Assurance (QA) and Quality Control (QC) measures ensures that experiments yield reliable and reproducible results.

### Pre-Lab Questions:

1. Why is a lab notebook considered a legal document?
2. How do SOPs, QA, and QC contribute to experimental consistency and reproducibility?
3. What are the potential consequences of poor lab documentation in a research or industry setting?
4. Why is proper donning and doffing of PPE important in a laboratory environment?

## Chapter Four

### Lab Notebook

In this Chapter, we're delving into the often underestimated yet immensely crucial tool in scientific endeavors – the **lab notebook**. It's not just about record-keeping; it's a cornerstone for companies seeking patents and preserving intellectual property. Now, think about companies such as Apple and Microsoft. Their success is intricately tied to the meticulous documentation of innovations in their lab notebooks. These notebooks are more than mundane records; they safeguard intellectual property and serve as a fortress against competitors trying to replicate proprietary information. Let's take a closer look at Apple. Suppose their research and development team is working on a groundbreaking software feature for the next iPhone or AI program. Every step of the innovation process, from coding algorithms to designing user interfaces, is meticulously recorded in their lab notebooks. This not only ensures accurate documentation but also plays a vital role in securing patents that protect their innovative creations.

Remember, lab notebooks are not mere record-keeping tools; they become legal documents establishing ownership of ideas or discoveries. This is especially crucial for companies seeking patents, as it proves the legitimacy of experiments and results. Consider a biotech company filing a patent for a groundbreaking medical device. In the event of a patent dispute, the inventors' lab notebooks become indispensable legal documents. They serve as evidence to establish ownership and demonstrate the originality of the invention. These notebooks can be presented in court to settle disputes, ensuring the company retains exclusive rights to its innovation.

In a biotechnology company, researchers conducting experiments on a new pharmaceutical product meticulously document each step in their lab notebooks, adhering to **Good Manufacturing Practices (GMPs)**. These notebooks play a crucial role during government regulatory audits. Proper documentation is essential for demonstrating compliance with GMPs, ensuring product quality, and meeting regulatory standards. Any lapses in documentation could lead to fines or legal consequences for the company.

In an academic research lab, a group of scientists exploring a novel scientific phenomenon follows a similar path. Each researcher maintains a lab notebook with consistent documentation following the **Standard Operating Procedure (SOP)**. SOP ensures that every researcher follows the same protocol, leading to consistency in results. **Quality Assurance (QA)** measures guarantee that each step is performed consistently, eliminating variations. **Quality Control (QC)** checks involve comparing current results with historical data, ensuring the reliability and reproducibility of the experiment.



**SOP (Standard Operating Procedure):** It's essentially the methods documented in the lab notebook but with a crucial difference. It represents a consensus protocol agreed upon by the research team – a standardized set of instructions that everyone follows.

**QA (Quality Assurance):** QA involves the consistent application of the SOP. It ensures that every step of the experiment is performed according to the agreed-upon protocol, maintaining a high level of consistency.

**QC (Quality Control):** QC involves double-checking the results against previous outcomes. This step ensures that the current results align with historical data, emphasizing the importance of consistency in laboratory work.

So, our lab notebooks are not just about recording experiments; they are powerful tools ensuring legal protection, regulatory compliance, and the reliability of scientific discoveries. Let's dive deeper into the intricate world of lab notebooks and their multifaceted roles.

Alright, class, let's dive into the intricacies of a well-structured **lab notebook**. It's not just a collection of random scribbles; it follows a meticulous format designed for clarity, traceability, and reproducibility of experiments. See



Figure 4.1 as reference:

First things first, we have the **cover page** – the face of the lab notebook. It provides crucial information like the class name, year, semester, instructor, school, and the researcher's name. This information is like a quick reference guide, helping us identify the notebook's ownership and context. Following the cover page, we have the **table of contents** on the first few pages. It's our roadmap, listing page numbers, titles, and dates for easy navigation through the notebook.

Now, let's break down the research sections:

**Purpose:** This sets the stage for the experiment, articulating what the researcher aims to achieve. For example, "The purpose of this section is to demonstrate how to recalibrate a pH meter." Clear purposes help us understand the goals and expectations. **Materials and Method:** The method section is a detailed account of the experimental procedure. It includes a list of materials required, specifying the location, catalog, and company of each item. This meticulous documentation ensures reproducibility, preventing the need to scramble for materials during the experiment. **Results:** This section contains observations, descriptions, or graphical representations of the outcomes. It might include images or graphs showcasing what the researcher observed during the experiment. **Discussion:** The discussion section provides an opportunity to convey additional information not covered in the materials and methods. It allows the researcher to explain any discrepancies or unexpected results. For instance, revealing that the buffer material used was expired, impacting the experiment's outcome.

Consistency takes center stage in the laboratory environment, where high-quality work is synonymous with reliability and uniformity. Whether we're documenting procedures in our lab notebooks, adhering to SOPs, ensuring QA, or conducting QC checks, the overarching goal is to cultivate a culture of trust in scientific practices. Our lab notebooks stand as a testament to our commitment to accuracy, honesty, and meticulous record-keeping. In conclusion, our lab notebook surpasses its role as a mere record-keeping tool; it transforms into a legal document safeguarding intellectual property and fostering a culture of consistency and trust. Drawing inspiration from industry giants, we recognize the pivotal role lab notebooks play in innovation and patent protection. As educators and researchers, instilling comprehensive lab notebook practices becomes imperative. It equips the next

generation of scientists with essential documentation skills and reinforces the ethical and responsible conduct of scientific endeavors.

### LAB ACTIVITY - PART 2 - Standard Operating Procedures for DONNING AND DOFFING

Donning and doffing are critical processes in a laboratory setting for ensuring the safety of individuals working with hazardous materials. These procedures are essential for preventing contamination and maintaining a sterile environment.

**Donning (Putting on):** Donning involves putting on personal protective equipment (PPE) such as lab coats, gloves, goggles, and masks. This creates a protective barrier between the individual and potentially harmful substances. Proper donning minimizes the risk of contamination, ensuring that individuals are shielded from exposure to chemicals, biological agents, or other hazardous materials. Following a standardized procedure for donning ensures compliance with safety regulations and guidelines, promoting a culture of safety within the laboratory.

**Doffing (Taking off):** Doffing is equally important to prevent cross-contamination. It ensures that any potentially contaminated PPE is removed carefully, minimizing the risk of spreading hazardous materials. Doffing is a key component of maintaining personal and environmental hygiene in the laboratory. It prevents accidental contact with hazardous substances after completing tasks. Doffing includes the correct disposal of used PPE, contributing to the proper management of laboratory waste and adhering to environmental and safety regulations.

Writing a Step-by-Step Protocol for Donning and Doffing:

Creating a Standard Operating Procedure (SOP) for donning and doffing ensures consistency and adherence to safety protocols. Here's a brief guide on how to write an SOP, with a focus on including putting away personal belongings, wearing lab coats, and disinfecting the lab bench:

#### Title and Introduction:

- Clearly state that the SOP is for the donning and doffing of PPE in the laboratory.
- Provide a brief introduction highlighting the importance of these procedures for safety.
- List the specific PPE required for the laboratory tasks.
- Include any additional materials such as disinfectants and storage spaces for personal belongings.

#### Donning Procedure:

- Detailed step-by-step instructions and the proper order for entering the lab.

#### Doffing Procedure:

- Detailed step-by-step instructions and the proper order for exiting the lab.

#### Final Checks:

- Include a checklist for individuals to ensure they have followed all steps correctly.

-----  
Here's a simple example for donning and doffing PPE in a kitchen environment:

#### Standard Operating Procedure (SOP)

**Title:** Donning and Doffing PPE for Kitchen Safety

**Purpose:** This SOP outlines the proper steps for donning and doffing personal protective equipment (PPE) to ensure safety and hygiene while working in the kitchen. Proper use of PPE minimizes contamination risks and promotes adherence to safety protocols.

#### Materials:

- Clean apron or chef's coat
- Gloves
- Hairnet or cap
- Goggles (if handling chemicals like cleaning agents)
- Mask (optional, based on kitchen guidelines)
- Disinfectant wipes or spray
- Storage area for personal belongings

## Donning Procedure

### 1. Preparation:

- Store personal belongings (e.g., phones, bags) in designated lockers or areas outside the kitchen.
- Wash hands thoroughly with soap and water for at least 20 seconds.

### 2. Putting on PPE:

- **Step 1:** Secure your hair with a hairnet or cap to prevent hair contamination.
- **Step 2:** Put on the apron or chef's coat, ensuring it covers your clothing completely. Fasten it properly.
- **Step 3:** If required, put on a mask, covering both nose and mouth. Adjust the fit.
- **Step 4:** Wear goggles if you are handling cleaning chemicals or tasks involving splashes.
- **Step 5:** Finally, put on clean gloves, ensuring they fit snugly over your hands.

### 3. Final Check:

- Verify that all PPE is securely worn and covers the appropriate areas.
- Disinfect your workstation with wipes or spray before starting work.

## Doffing Procedure

### 1. Preparation:

- Before removing PPE, wipe down surfaces you worked on using disinfectant to avoid cross-contamination.

### 2. Removing PPE:

- **Step 1:** Carefully remove gloves by peeling them off inside out, without touching the skin. Dispose of them in a trash bin.
- **Step 2:** Remove the goggles, handling them by the arms. Clean and store them if reusable.
- **Step 3:** If wearing a mask, remove it by the ear loops without touching the front. Dispose of or store as per guidelines.
- **Step 4:** Unfasten and remove the apron or chef's coat, avoiding contact with its outer surface. Place it in a laundry bin or designated area.
- **Step 5:** Remove the hairnet or cap and dispose of it if disposable.

### 3. Post-Doffing Hygiene:

- Wash hands thoroughly with soap and water for at least 20 seconds.

### 4. Final Check:

- Ensure all PPE has been disposed of or stored properly.
- Confirm that personal belongings remain outside the kitchen area to maintain hygiene.

---

## Post-Lab Questions:

1. What are the key elements that must be recorded in a lab notebook to ensure proper documentation?
2. How does maintaining a well-structured lab notebook help in securing patents and protecting intellectual property?
3. Describe an instance in which failing to follow SOPs, QA, or QC measures could impact research outcomes.
4. What challenges did you encounter while writing the SOP for donning and doffing PPE, and how did you address them?

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## 5: Measurements and Unit Conversion

### Learning Objectives

- Students will explore the importance of standardized units, accuracy, precision, and unit conversions in scientific research.
- Students will recognize how miscalculations can impact experimental outcomes and reinforce the significance of the metric system

### Pre-Lab Questions

1. What is the difference between accuracy and precision, and why do both matter in scientific measurements?
2. Why is the metric system preferred in scientific research over the empirical system?
3. How does the base 10 structure of the metric system simplify unit conversions?
4. What are the potential real-world consequences of incorrect unit conversions in biotechnology or other scientific fields?
5. How can you ensure that your measurements are both accurate and precise in a laboratory setting?

---

## Chapter Five

### Measurements and Unit Conversions

Alright, let's understand why precision and accuracy in measurements are absolutely crucial for ensuring the reliability of our experiments. In this chapter, we'll explore the significance of standardized units, accuracy, precision, and unit conversions. Now, why is avoiding ambiguity in measurements so essential? It all comes down to **standardized units** – a set of agreed-upon measurements used universally. These units ensure clarity and consistency in communication. Let me illustrate this with a scenario: Imagine you tell your grandparents that you're as tall as your jeep. Now, this vague comparison can lead to misunderstanding because the interpretation of "jeep" varies, like a kid's toy jeep. Instead, using a standardized unit like meters or feet provides a precise and universally understood measurement. When we respond to inquiries about height with vague comparisons, it highlights the need for standardized units to convey accurate information. So, as we dive into the details, let's remember the importance of standardized units, accuracy, and precision in our scientific measurements to ensure our experiments are reliable and trustworthy.

First, let's clarify these concepts. **Accuracy** refers to how closely a measurement aligns with the true value, like hitting the bullseye when aiming for a target. On the other hand, **precision** is all about consistency – hitting the same spot every time, even if it's not the bullseye. Ideally, in scientific experiments, we aim for both accuracy and precision. It's crucial to understand and apply these concepts because a precise yet consistently inaccurate experiment can lead to misleading results.

Now, the choice between the **empirical** and **metric** systems plays a significant role in scientific research. While the **empirical system** is familiar in the United States, the **metric system** offers simplicity in **unit conversion**. Converting units in the metric system involves a straightforward shift of the decimal point. For example, converting 100 centimeters to meters requires moving the decimal two places to the left: 100 cm = 1.00 meters. In contrast, the empirical system involves more complex calculations when converting inches to feet to yards to miles. Miscalculations or oversights in units can lead to catastrophic consequences in various industries. In fields like aerospace, incorrect fuel measurements in a rocket launch could result in mission failure or accidents. The chapter emphasizes the advantages of the metric system, particularly in experiments where precision is crucial.

Unit conversion is a fundamental skill in scientific endeavors, and the metric system simplifies this process. Shifting the decimal to the left for larger units and to the right for smaller units is a straightforward approach. Converting 400 milliliters (mL) to liters (L) involves moving the decimal point three places to the left: 400 mL = 0.4 L. Practical examples, such as converting milliliters to liters and micrometers to meters, illustrate the ease of unit conversion in laboratory settings.

In our biotechnology adventures, measuring the mass of substances becomes a breeze when we use grams (g) as our unit. This not only ensures consistency but also makes our measurements easily reproducible. Our trusty instruments, like balances for mass in grams and graduated cylinders for volume in milliliters, are tailored to the metric system, ensuring accurate data collection. Remember, in the metric system, we have base measurements like the meter for distances, the liter for volume, and the gram for

mass. Every measurement should include both a numerical value and a unit, showing its magnitude relative to the base unit. In our biotechnology labs, we stick to specific metric units such as meters (m), grams (g), and liters (L), making experiments comparable and reproducible.

| Base Units of Measure |           |                                     |              |
|-----------------------|-----------|-------------------------------------|--------------|
| Length:               | meter (m) | Time:                               | second (s)   |
| Mass:                 | gram (g)  | Temperature:                        | Celsius (°C) |
| Volume:               | Liter (L) |                                     |              |
| Kilo-                 | (k)       | 1,000                               | $10^3$       |
| Hecto-                | (h)       | 100                                 | $10^2$       |
| Deca-                 | (da)      | 10                                  | $10^1$       |
| One                   |           | 1                                   | $10^0$       |
| Deci-                 | (d)       | 0.1 (= $\frac{1}{10}$ )             | $10^{-1}$    |
| Centi-                | (c)       | 0.01 (= $\frac{1}{100}$ )           | $10^{-2}$    |
| Milli-                | (m)       | 0.001 (= $\frac{1}{1000}$ )         | $10^{-3}$    |
| Micro-                | (μ)       | 0.000001 (= $\frac{1}{1,000,000}$ ) | $10^{-6}$    |

As you get a larger unit, you shift your decimal to the left ←  
 As you get smaller unit, you shift your decimal to the right →

Now, what makes the metric system even more amazing is its base 10 scaling (Figure 5.1) This simplifies unit conversion, as every unit is just a power of 10 away from the others. Converting from centimeters to meters? Easy! Just move the decimal point two places to the left. For example, 100 cm = 1.00 meters. This straightforward scaling ensures a smooth conversion process, a crucial aspect in the precision of our biotechnological measurements. Practical insights, like shifting the decimal point, further reinforce the simplicity of the metric system.

In conclusion, concepts like standardized units, accuracy, precision, and unit conversion are the building blocks of successful experiments. The metric system is our reliable ally in biotechnology labs, providing a standardized framework and fostering a culture of precision in our scientific inquiry. So, as we navigate the intricate landscape of research, let's embrace the metric system for meticulous record-keeping and ensure accurate and meaningful discoveries.

### LAB ACTIVITY - PART 1 - UNIT CONVERSION

To reinforce your understanding of unit conversions, I encourage you to engage in regular practice. Proficiency in converting units is a fundamental skill in various scientific disciplines, and consistent practice will enhance your confidence and accuracy.

1. 50mm to: \_\_\_\_\_ cm
2. 50cm to: \_\_\_\_\_ km
3. 20m to: \_\_\_\_\_ μm
4. If I am 1.78m tall, how many cm am I?
5. Convert 5 kilometers to meters.
6. Express 3.5 liters in milliliters.

Convert 200 grams to kilograms.

-----

#### Summary

Now that you have completed the lab activity, take a moment to reflect on the concepts learned. This post-lab section will help you consolidate your understanding and evaluate how effectively you applied unit conversion skills.

#### Post-Lab Questions

1. What challenges did you encounter when converting between units, and how did you overcome them?
2. How did using the metric system simplify your calculations compared to conversions in the empirical system?
3. Describe a real-world scenario in biotechnology or healthcare where precise unit conversions are essential.
4. If you were explaining the importance of accuracy and precision to a new scientist, how would you illustrate the difference?
5. Reflect on a measurement you took during the lab—was it accurate, precise, or both? How could you improve your technique?

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## 6: Significant Figures and Standard Mass

### Learning Objectives

- Understand the importance of significant figures in scientific measurements.
- Learn how to properly read volume using a meniscus.
- Compare the precision of different measuring tools.
- Verify the accuracy of measuring instruments using standard masses.
- Perform serological pipetting using TC, TD, and reverse pipetting methods.

### Definition: Term

- **Significant Figures:** Define and practice identifying significant figures in various measurements.
- **Meniscus Reading:** Explain how to properly read liquid volume in a graduated cylinder or pipette.
- **Precision vs. Accuracy:** Differentiate between these terms in the context of measurement tools.
- **Absolute and Percent Error:** Learn how to calculate and interpret these errors when using analytical balances and pipettes.
- **Types of Serological Pipettes:** Compare TC and TD pipettes and understand when to use each.

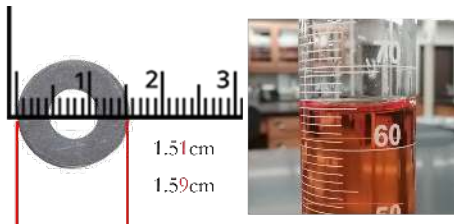
### Pre-Lab Questions

1. Why are significant figures important in scientific measurements?
2. How does the meniscus affect the accuracy of volume measurements?
3. What is the difference between absolute error and percent error? Provide an example.
4. How do TC and TD pipettes differ, and why does it matter in liquid handling?
5. Why is it important to verify an analytical balance using a standard mass?

## Chapter Six

### Significant figures and Standard mass

Today, we're diving into the intricacies of measurements, focusing specifically on significant figures, the accuracy of measuring tools, and troubleshooting techniques to ensure precision in our scientific endeavors. Let's start with **significant figures**, which are crucial for accurately representing the precision of our measurements. They tell us the number of reliably known digits in a measurement. Consider using a ruler to measure an object where the true value lies between 1.5 cm and 1.6 cm (Figure 6.1). If our measurement falls between these two hash marks, the significant figures would include the digit assumed between them, ranging from 1.51 cm to 1.59 cm. Each of these digits represents a reliably known part of the measurement, emphasizing the importance of understanding significant figures to capture precision accurately.



When it comes to reading volume, especially with tools like a graduated cylinder or serological pipette, we encounter the **meniscus** (Figure 6.2). The meniscus, the bottom of the concave shape formed by the liquid, is considered the true volume. This practice accounts for the curvature caused by the cohesion and adhesion properties of water, providing a reliable significant figure for volume measurement.

Moving on to the accuracy of measuring tools, let's consider a micropipette. Such a tool offers more significant figures and precision compared to a serological pipette or a graduated cylinder. However, it's important to note that relying solely on the device may not guarantee accurate measurements. Human error and equipment limitations can impact the consistency of our

measurements. Imagine using a micropipette with high precision to measure 1.000 mL. If there's a human error or a malfunction in the pipette, the volume measured might be 1.005 mL.

An effective method for ensuring equipment reliability involves the verification of instruments using a **standard mass**—a known and accurate weight. Let's consider an example with an electronic balance: if you weigh a standard mass of 100g and the balance reads 100.650g, the **absolute error** is calculated as  $(100.000\text{g} - 100.650\text{g})$ , resulting in a deviation of -0.650g from the expected value. Subsequently, let's say you measure 4.3 mL of water, but the balance reads 4.85 grams. Without applying the absolute error, the **percent error** is calculated as  $(4.85\text{g} - 4.3\text{g})/4.3\text{g} * 100$ , resulting in a high 13% error. This high percentage might initially lead to attributing the error to human error or the inaccuracy of the micropipette. However, if you account for the absolute error by subtracting it, the actual weight becomes 4.20g. Now, with the adjusted value, the percent error is recalculated as  $(4.3\text{g} - 4.20\text{g})/4.3\text{g} * 100$ , yielding a lower 2% error. This adjustment emphasizes the importance of considering absolute error, as it can significantly impact the perceived accuracy of measurements and guide decisions on whether adjustments or calibration are necessary for the device to ensure precise and reliable results.

### LAB ACTIVITY - PART 2 - Verification of Analytical Scale

In order to ensure accurate measurements during your practical work, it's crucial to verify the absolute value of your analytical scales. Follow these steps to assess the accuracy of each scale using standard masses:

1. Assign a unique letter code to each analytical scale. This code will be used to identify and record the measurements in your lab notebook.
2. Obtain standard masses of 1g, 5g, 10g, 50g, 100g, and 200g. These masses will serve as reference points for calibration.
3. Place each standard mass individually on the designated scales. Record the displayed mass reading for each standard weight on the analytical scale.
4. Subtract the theoretical standard mass by the recorded mass reading for each measurements. The absolute value represents the difference between the displayed value and the standard mass.
5. Create a dedicated section in your lab notebook for each scale's verification. Clearly document the letter code of the scale, the standard mass used, the recorded mass reading, and the calculated absolute value.
6. (Optional) Repeat the process for each standard mass on the same scale to ensure precision. If there are discrepancies, note them down and consider repeating the verification.

These information will be valuable for future assessments, quizzes, or practical work where you can choose the scale that aligns with your personal preference.

### LAB ACTIVITY - PART 3 - Serological pipetting

Serological pipettes play a crucial role in precisely measuring and dispensing liquids in various biotechnological applications. **Serological pipettes** come in two main types: "to contain" (TC) and "to deliver" (TD). **TC pipettes** are designed to hold a specified volume when filled to the capacity mark. However, due to liquid adherence, they may not deliver that exact amount when poured out. On the other hand, **TD pipettes** are marked differently to ensure precise delivery of the specified amount under specific conditions. Choosing the correct pipette, whether TC or TD, is critical for accurate liquid handling.

One fascinating technique we use with serological pipettes is called "**Reverse Pipetting**." Unlike conventional methods where liquid is expelled by the pipette's blow-out mechanism, reverse pipetting involves aspirating more liquid than needed and then dispensing the excess. This technique is invaluable when precision is paramount and avoiding the introduction of contaminants is crucial.

Now, let's walk through the steps of using serological pipettes in the lab:

1. Selecting the Appropriate Serological Pipette and Tips:
  1. Choose the appropriate serological pipette based on the desired volume to be dispensed. Also, ensure that compatible 25mL serological pipette tips are available.
2. Aspirating the Liquid to the Desired Volume:
  1. **TD Delivery:** Draw up exactly 5mL of liquid using the serological pipette. Dispense the 5mL volume into three different cups.
  2. **TC Delivery:** Draw up 6mL of liquid using the serological pipette. Dispense 5mL the 5mL volume into three different cups.
  3. **Reverse Pipetting:** Aspirate up to 20mL of liquid using the serological pipette. Dispense 5mL at a time into three different cups.



**Challenge:**

1. Place an empty cup on the balance and tare it to zero.
  2. Applying TD, TC, or Reverse Pipetting Method:
  3. Using the appropriate method, deliver the following solutions into the tared cup
    1. 3mL water
    2. 5mL water
    3. 4mL water
    4. 10mL water
  4. Record the time taken for each delivery and determine your percent error
    1.  $\%Error = ((Measured\ Volume - Expected\ Volume) / Expected\ Volume) * 100\%$
  5. Continue timing yourself with different volumes of water and determine the percent error.
- 

**? Post-Lab Questions:**

- Based on your analytical balance verification, how accurate were your measurements? Were there any patterns in the errors?
- How did adjusting for absolute error impact the percent error in your calculations?
- Which pipetting method (TD, TC, or reverse pipetting) resulted in the lowest percent error? Explain why.
- If given a choice of measurement tools (micropipette, serological pipette, graduated cylinder), which would you choose for the most precise measurement and why?
- How would inaccurate measurement tools affect experimental results in a real-world biotechnology application?
- What challenges did you face in using the measuring tools, and how did you overcome them?
- What did you learn about precision and accuracy that you didn't know before?
- How can these concepts be applied in professional laboratory settings?

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## 7: Water - The Solvent of Life

### ? Pre-Lab Questions:

1. What are some everyday examples where you see water acting as a solvent?
2. Why do you think oil and water do not mix?
3. Have you ever noticed that a glass of water left out overnight has bubbles on the sides? What do you think causes this?
4. What do you predict will happen when sugar, salt, and oil are added to water? Explain why you think so.
5. How do you think temperature affects water's ability to dissolve substances?
6. Define polarity and explain why water is considered a polar molecule.
7. What is surface tension, and how does cohesion contribute to it?
8. What is the pH scale, and how does water help regulate pH in biological systems?

### Chapter Seven

#### Water -- The Solvent of Life

Let's dive into the fascinating world of water! We all know **water** is essential for life, but did you know it's so much more than just a thirst-quencher? Today, we're going to explore the amazing properties of water. Water has an incredible ability to dissolve stuff—it's what we call being the "**solvent of life**." This all comes down to water's **polar** nature, which means it has a slightly positive end and a slightly negative end. Picture it like a tiny magnet! Because of this polarity, water can grab onto other molecules and break them apart. For example, when you make a cup of tea and add sugar to hot water, the sugar dissolves because water's polarity attracts the sugar molecules, sweetening your drink. Similarly, when salt is added to water, its molecules break apart because the water molecules surround and separate the sodium and chloride ions, causing the salt crystals to vanish into the water. This process is why the ocean is salty—because over time, minerals like salt dissolve in water, making it salty. But if something doesn't have these charges, like **nonpolar** oils, water won't mix with it. That's why oil and water always separate, just like they say, "like dissolves like."

Furthermore, water's ability to absorb and release heat plays a crucial role in regulating temperature, thanks to its **hydrogen bonding**. When heat is applied to water, it's absorbed as the hydrogen bonds break, allowing water molecules to move more freely and increasing the temperature. However, water's **cohesive** forces work to moderate this temperature rise by **quickly forming new bonds** between neighboring molecules, which slows down the increase in heat. This means water can resist temperature changes, which is essential for maintaining stable conditions in both living organisms and the environment. Let's relate this to everyday situations for better understanding. Imagine spending a day at the beach. As the sun beats down, the sand gets scorching hot, but the ocean nearby stays relatively cool. This is because water absorbs heat slowly from the sun during the day and releases it slowly at night, helping to keep the temperature stable. Similarly, inside our bodies, water helps regulate temperature. When we exercise and our bodies heat up, sweat glands produce sweat, which evaporates from our skin, taking heat away from our bodies and cooling us down. So, whether it's at the beach or during exercise, water's ability to manage heat is vital for our comfort and well-being.

Let's delve into how water's cohesion and adhesion showcase its extraordinary characteristics. **Cohesion** refers to water molecules' tendency to stick together, thanks to hydrogen bonding. This cohesive force allows water to form droplets, create surface tension, and move against gravity through narrow spaces, like the capillary action observed in plants. For instance, think of a raindrop clinging to a leaf or a spider effortlessly walking on the surface of a pond without sinking. These examples highlight water's cohesive properties, where its molecules stick together, forming distinct droplets or supporting small creatures on the water's surface. Conversely, **adhesion** demonstrates water's attraction to other substances. For example, when you dip a paper towel into water, the water climbs up the towel against gravity due to adhesion. So, whether it's forming droplets on a leaf or climbing up a paper towel, water's interplay of cohesion and adhesion illustrates its remarkable versatility and importance in various everyday and biological processes.

Now, let's explore water's role in regulating pH. The **pH scale**, ranging from 0 to 14, helps us measure how acidic or alkaline a solution is. It's based on the concentration of hydrogen ions ( $H^+$ ) and hydroxide ions ( $OH^-$ ) in the solution. Solutions with a pH below 7 are **acidic** because they have more hydrogen ions, while those above 7 are **alkaline** or **basic** due to higher concentrations of hydroxide ions. Water itself is **neutral**, with a pH of 7, because it maintains an equal balance of hydrogen and hydroxide ions through autoprotolysis. Let's consider some examples to understand this better. Lemon juice is acidic because it contains citric acid.

When you add lemon juice to water, its pH decreases because of the hydrogen ions from the citric acid. These hydrogen ions can be damaging to living tissues; for instance, if a strong acid comes into contact with your skin, the excess  $H^+$  ions can bind to the negatively charged phospholipid bilayer of your cells, causing a burning sensation. On the other hand, baking soda (sodium bicarbonate) is alkaline. When mixed with water, it releases hydroxide ions, increasing the pH of the solution and making it more basic. Basic solutions can feel slippery because of the presence of hydroxide ions, which repel against the negative charge of your phospholipid. However, it's crucial to understand that basic solutions can also be harmful. For instance, if a strong base touches your skin, it can dissolve the skin by reacting with the lipids in it. This occurs because, as mentioned earlier, like-dissolves-like, and the hydroxide ions in basic solutions can disrupt the lipid barrier of the skin.

Now, onto the math part. Understanding the pH scale involves knowing that it's logarithmic. This means that each unit change in pH represents a **ten-fold** difference in hydrogen ion concentration. For instance, a solution with a pH of 3 has 100 times more hydrogen ions than one with a pH of 5. This logarithmic relationship helps us grasp the significant impact that small changes in pH can have on chemical reactions, biological processes, and environmental stability.

To wrap up, water truly stands out as a wonder of nature, showcasing a complex array of properties and functions crucial for the sustenance and development of life. Whether it's acting as the universal solvent, regulating temperature, displaying cohesive and adhesive behaviors, or impacting pH levels, water influences every aspect of our lives in profound ways. As we continue to explore the intricacies of this simple yet extraordinary molecule, we unveil layers of complexity that highlight its irreplaceable role in the intricate web of life.



*"bw water drop" by Muffet is licensed under CC BY 2.0.*

### ? Post-Lab Questions

- What happened when you added sugar, salt, and oil to water? How did the results compare with your predictions?
- Describe any patterns or unexpected results you observed. What might explain these findings?
- How does water's polarity help explain why certain substances dissolve while others do not?
- In what ways did you see cohesion and adhesion at work in the lab?
- How do your findings about water's heat absorption relate to real-world scenarios, such as climate regulation or body temperature control?
- If you were to design an experiment to test water's solvent abilities with different temperatures, how would you set it up?
- What role do you think water plays in maintaining pH balance in living organisms?
- How do the properties of water make it essential for life on Earth?
- What was the most surprising or interesting thing you learned about water?
- How has this lab changed the way you think about water's role in everyday life?

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## 8: Micropipetting

### Learning Objectives

- Students will understand the purpose of micropipetting, its applications, and the principles of accurate liquid measurement. This section prepares students for precise pipetting techniques and error analysis

### Definition: Term

- Why Micropipettes Matter:
  - Micropipettes allow accurate and precise measurement of microliter volumes in laboratory experiments.
  - They are commonly used in molecular biology, enzymatic assays, and cell culture work.
- Types of Micropipettes:
  - P-1000: Measures 100-1000  $\mu\text{L}$  (used for buffers and media)
  - P-200: Measures 20-200  $\mu\text{L}$  (used for PCR and DNA/RNA quantification)
  - P-20: Measures 2-20  $\mu\text{L}$  (used for precise enzyme or reagent additions)
- Reading the Volume Display:
  - Understand how to read the digital display for different micropipette models.
  - Identify correct settings to ensure accurate volume transfer.
- Correct Pipetting Technique:
  - Press plunger to the first stop before submerging the tip in the liquid.
  - Slowly release the plunger to aspirate the liquid while avoiding bubbles.
  - Dispense the liquid smoothly against the side of the receiving vessel.
- Common Errors to Avoid:
  - Using the wrong micropipette for a specific volume.
  - Pipetting too fast, causing air bubbles.
  - Not changing tips between different samples (risk of contamination).
  - Mishandling or dropping the micropipette.

### Pre-Lab Questions:

1. Why is accuracy important when using micropipettes?
2. Which micropipette would you use to measure 150  $\mu\text{L}$ ? 15  $\mu\text{L}$ ? 750  $\mu\text{L}$ ?
3. Describe the difference between the first and second stop of the micropipette plunger.
4. What are some potential errors that could affect pipetting accuracy? How can you prevent them?
5. Why is it important to record percent error during pipetting verification?

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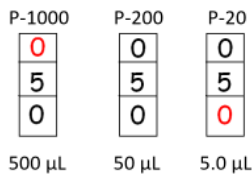
## Chapter Eight

### Micropipetting

Accurately getting the measurements is super important, especially when we're dealing with really tiny amounts of liquids. **Micropipettes** are important in the lab because they help us do this job with extreme accuracy. When we need to measure tiny volumes of liquids, usually in the microliter range (that's like really, really small amounts), micropipettes are our go-to tools. Unlike regular pipettes where you have to withdraw and dispense liquids manually, micropipettes are way better. They use mechanisms to withdraw and dispense out liquids precisely, leaving very little room for error. Let's learn all about micropipettes - how they work and the best ways to use them in our experiments.

Operating a micropipette may appear simple at first glance, but truly mastering it requires careful attention to detail and plenty of practice. Micropipettes are available in various sizes to accommodate different volume ranges, commonly known as P-1000, P-200, and P-20, which are widely used in laboratories. Each micropipette is designed for specific volume ranges, and it's crucial to

choose the appropriate one based on the volume of liquid you need to measure to ensure accurate results. For instance, in a molecular biology lab, you might utilize a P-1000 micropipette for transferring larger volumes of buffer or media, a P-200 for PCR reactions, and a P-20 for precisely adding enzyme or primer solutions. So, when selecting a micropipette, consider the volume you're working with. The P-200 micropipette is typically used for volumes ranging from 20  $\mu\text{L}$  to 200  $\mu\text{L}$ . This range makes it suitable for various applications, such as dilutions, sample transfers, and DNA/RNA quantification. On the other hand, the P-20 micropipette is designed for even smaller volumes, typically ranging from 2  $\mu\text{L}$  to 20  $\mu\text{L}$ . This micropipette is ideal for tasks requiring extreme precision, such as PCR setup, restriction enzyme digestion, and pipetting small aliquots of expensive reagents. So, when selecting a micropipette, consider the volume you're working with and choose the appropriate one to ensure accurate measurements. If you need to transfer 500  $\mu\text{L}$  of a sample, opt for the P-1000 micropipette. This micropipette is capable of measuring volumes between 100  $\mu\text{L}$  and 1000  $\mu\text{L}$ , providing the accuracy and precision needed for your measurements.



Micropipettes also come equipped with display windows that show the volume being measured, and knowing how to interpret these displays is key for accurate pipetting. See Figure 8.1. Understanding the calibration markings on the micropipette is crucial for precise pipetting. These markings indicate different volume increments, so becoming familiar with them is essential.

For instance, let's take a look at the display window of a P-1000 micropipette. You'll notice red numbers indicating thousands, followed by smaller lines representing hundreds, tens, and maybe ones. Similarly, the P-200 and P-20 micropipettes have their own display formats tailored to their specific volume ranges.

Let's explore the withdrawal and dispensing technique. When **withdrawing** liquid, it's vital to press the plunger down to the first stop. This action creates a vacuum before you put the tip into the liquid. Doing this ensures that you aspirate the desired volume accurately without introducing any pesky air bubbles. It's important to take it slow when you're drawing up the liquid to prevent those air bubbles from forming, as they can mess with the accuracy of your measurements. Now, when it comes to **dispensing** the liquid, you'll want to press the plunger down to the first stop, just like before. Then, you'll smoothly and steadily release it. To minimize any liquid sticking to the tip, it's a good idea to **touch the side of the vessel** as you dispense. Another handy technique to remember is **airdropping**. This involves expelling a tiny bit of liquid from the tip after you've dispensed the main volume. Airdropping helps prevent any contamination and ensures that you deliver the volume precisely where you need it. It's crucial to watch out for air bubbles in the tip because they can throw off your volume measurements. If you spot any bubbles, you'll need to expel the liquid and start the pipetting process again.

Some best practices and considerations to keep in mind:

- **Take it Slow:** Avoid the temptation to withdraw or dispense liquids too quickly. Doing so can introduce inaccuracies and cause inconsistent results. For example, in PCR reactions, rapid dispensing might mess with the efficiency of the reaction, leading to unpredictable amplification outcomes.
- **Fresh Tips:** It's crucial to always use new tips for each transfer. This helps prevent any sneaky cross-contamination between samples. Imagine you're doing cell culture experiments - using contaminated tips could mean unintentionally contaminating your cell cultures and throwing off your experimental results.
- **Calibration:** Make sure to regularly calibrate or at least verify your micropipettes. This ensures that they're measuring volumes accurately and precisely. Calibration is like giving your micropipette a tune-up, making sure it's working correctly and delivering the right volumes every time. And as you know, reliable experimental outcomes depend on accurate measurements.
- **Handle with Care:** Treat your micropipettes with the utmost care. They're delicate instruments with sensitive components. Mishandling them could lead to inaccuracies in your volume measurements and compromise your experimental results.

Let's explore some real-world applications. In enzymatic assays, getting the volumes of substrates and enzymes just right is crucial for figuring out reaction kinetics and enzyme activity. Accurate micropipetting ensures that the right amounts of reactants are mixed, leading to reliable assay results. For example, when we're studying an enzyme involved in DNA replication, making sure we measure enzyme and substrate volumes precisely helps us determine enzyme kinetics and how efficiently DNA gets copied.

PCR is a super common technique in molecular biology, and it relies heavily on precise pipetting of things like DNA templates, primers, and building blocks called dNTPs. If we mess up the pipetting, it can mess up the whole reaction, leading to wonky results. For instance, when we're amplifying a specific DNA sequence for genetic analysis, making sure we measure PCR reagents accurately ensures that we get a reliable and consistent amplification of our target DNA. In cell culture experiments, we need to prepare growth media and reagents with extreme care because accurate pipetting is crucial for keeping our cells happy and healthy. If we're not careful with our pipetting, we can introduce variability in the conditions our cells are growing in, which can mess with our results. For example, when we're growing mammalian cells for drug screening assays, making sure we measure media components precisely ensures that our cells are always in the best conditions for our experiments.

In conclusion, mastering the art of micropipetting is absolutely essential for success in the laboratory. By understanding how micropipettes work, following best practices, and practicing our skills, scientists can ensure that their experiments are precise, accurate, and reproducible. Whether we're doing enzymatic assays, PCR reactions, or cell culture experiments, being really good at micropipetting is key to getting reliable and meaningful results in scientific research.

## LAB ACTIVITY - PART 1 - Verification of Micropipettes

**Objective:** To verify the accuracy of micropipettes using an electronic balance.

**Notes:** Keep accurate records of the % error for each micropipette, as this information will be crucial for selecting the appropriate micropipette during a skill evaluation quizzes and practicum.

### Materials:

- Micropipettes (P-1000, P-200, P-20)
- Tips compatible with each micropipette
- Distilled water
- Electronic balance
- Weighing boats or beakers
- Laboratory notebook and pen

### Procedures:

1. Make sure your micropipettes are clean and ready to use.
2. Have your electronic balance set up and turned on.
3. Tare a measuring cup.
4. Measure with P-1000:
  - Use the P-1000 micropipette to measure water three times for each volume: 1000  $\mu$ L, 500  $\mu$ L, and 100  $\mu$ L.
  - Write down the mass of each measurement from the electronic balance.
  - Calculate the average % error for each volume using the formula: % Error =  $[(\text{Measured Mass} - \text{Theoretical Mass}) / \text{Theoretical Mass}] \times 100$ .
5. Measure with P-200:
  - Measure water five times for each volume: 200  $\mu$ L, 100  $\mu$ L, and 20  $\mu$ L.
6. Measure with P-20:
  - Measure water fifty times for each volume: 20  $\mu$ L, 10  $\mu$ L, and 2  $\mu$ L.
7. Grab tape, stick it onto the micropipette, and then write down the percent error on the tape. Include the date and your initial
8. Continue with other micropipettes, if any
9. Dispose of any used tips and water properly.
10. Wipe down your micropipettes and electronic balance to keep them clean.

Figure 8.2: Examples of how your data should be collected

|         | Theoretical (g) | Actual (g) | Percent error |
|---------|-----------------|------------|---------------|
| p1000 A | 3.000           | 3.070      | 2.33          |
|         | 1.500           | 1.520      | 1.33          |
|         | 0.300           | 0.321      | 7.00          |

Fill out the tables below:

### p1000 Micropipette:

| Theoretical weight (g) | Actual weight (g) | Percent error (%) |
|------------------------|-------------------|-------------------|
| 3.000 g                |                   |                   |
| 1.500 g                |                   |                   |
| 0.300 g                |                   |                   |

### p200 Micropipette:

| Theoretical weight (g) | Actual weight (g) | Percent error (%) |
|------------------------|-------------------|-------------------|
| 1.000 g                |                   |                   |
| 0.500 g                |                   |                   |
| 0.100 g                |                   |                   |

### p20 Micropipette:

| Theoretical weight (g) | Actual weight (g) | Percent error (%) |
|------------------------|-------------------|-------------------|
| 1.000 g                |                   |                   |
| 0.500 g                |                   |                   |
| 0.100 g                |                   |                   |

### Summary to show in the PowerPoint Slide

Use a micropipette and **verify** their accuracy via electronic balance

- Remember: 1mL of water = 1g of water

#### • Instruction

- Measure p1000 **three (3)** times for each volume, determine average % error
  - 1000uL, 500uL, and 100uL
- Measure p200 **five (5)** times for each volume, determine average % error
  - 200uL, 100uL, and 20uL
- Measure p20 **fifty (50)** times for each volume, determine average % error
  - 20uL, 10uL, and 2uL
- Grab a tape and write down the percent error, date, and initial for each micropipette

### LAB ACTIVITY - PART 2 - Skill Evaluation 1 (20pts)

During this skill evaluation, it's crucial to stress the significance of **efficiency** and **time management**. In laboratory settings, **time is valuable**, and being able to complete tasks quickly without compromising accuracy is essential for productivity.

**Objective:** To assess students' proficiency in accurately measuring volumes and transferring liquids into conical tubes and microcentrifuge tubes within a specified time frame.

#### Materials:

- Micropipettes and tips
- Serological pipettes and tips
- Mass standard
- Conical tubes
- Microcentrifuge tubes

- Liquid samples (red, blue, yellow, green)
- Timer
- Grading rubric

### Procedure

1. Begin by using the mass standard to determine the absolute error for the scale being used. Calculate the absolute error: \_\_\_\_\_ g
2. Set up a workstation with all necessary materials arranged neatly and within reach.
3. Preweigh a Conical tube and a Microcentrifuge tube
4. Add the specified volumes of each liquid to the corresponding tubes:
  1. Conical tubes: 6 mL red, 6 mL blue, 8 mL yellow, 30 mL green
  2. Microcentrifuge tubes: 600  $\mu$ L red, 300  $\mu$ L blue, 80  $\mu$ L yellow, 20  $\mu$ L green
5. Accurately complete the task within the designated time limit of 3 minutes.

### For TA or Instructor only:

6. Afterward, post-weigh both the Conical tube and the Microcentrifuge tube.
7. Measure the weight of the final solutions and calculate the percent error for both solutions in the Conical tube and the Microcentrifuge tube.
8. Accumulate the average percent error from both solutions, which will be subtracted from their score of 20 points.
  1. For instance, if their percent error is 8% from the Conical tube and 2% from the Microcentrifuge tube, the average percent error (used for grading purposes) will be 5%, resulting in a total of 15 out of 20 points.

### Extra Credits

- Award 2 extra credit points to any student who achieves a percent error of less than 1% on either the Conical tube and/or Microcentrifuge tube. Students may receive up to 4 extra points if they obtain under 1% error for both tests.

### Summary to show in the PowerPoint Slide

**Instructions:** Students will be given 4 minutes to fill their conical tube and microcentrifuge tube in front of their instructor or TA. Time will start when the students picked up any equipment. After 4 minutes, students must put all equipment down otherwise 5 points will be deducted. Students may have up to two tries. Extra credit for students who can get under 1% error for each task.

1. Pre-weigh a conical tube and a microcentrifuge tube
2. Add the following to the conical tube (Theoretical weigh ~ 50g)
  - 6mL - Red
  - 6mL - Blue
  - 8mL - Yellow
  - 30mL - Green
3. Add the following to the microcentrifuge tube. (Theoretical weigh ~ 1g)
  - 600 $\mu$ L - Red
  - 300 $\mu$ L - Blue
  - 80 $\mu$ L - Yellow
  - 20 $\mu$ L - Green
4. Weigh the conical tube and microcentrifuge tube, and determine the difference in weight, and then the percent error. The average percent error will be deducted from 20 points. (i.e., 1% error is -1pts.)

### ? Post-Lab Questions

1. Based on your percent error results, which micropipette size were you most accurate with? Why?
2. What adjustments could you make to improve your pipetting accuracy?
3. How does maintaining proper pipetting technique contribute to successful experiments in research and clinical settings?
4. If your percent error was consistently high, what steps would you take to troubleshoot and correct it?
5. What key takeaways did you learn from this lab that will help in future experiments

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## 9: Dilutions and Spectrophotometry

### Learning Objectives

- Understand the concept and purpose of serial dilution.
- Learn how to perform a 2-fold serial dilution accurately.
- Recognize the role of spectrophotometry in measuring solution concentrations.
- Identify the function of a blank in spectrophotometric analysis.
- Predict how dilution affects absorbance readings in a spectrophotometer.

### Note

Before coming to the lab, read the assigned chapter on dilutions and spectrophotometry. Pay close attention to:

- The mathematical principles behind serial dilution.
- How dilution factors are calculated.
- The operation and principles of spectrophotometry.
- The relationship between absorbance and concentration (Beer's Law).

### Pre-Lab Questions

1. Why is serial dilution an important technique in laboratory science?
2. If you begin with a 0.5M solution and perform a 1:2 dilution three times, what will be the final concentration?
3. How does a spectrophotometer measure concentration, and why do we use a blank solution?
4. Predict how the absorbance values will change as the concentration decreases in your dilution series.

## Chapter Nine

### Dilutions and Spectrophotometry

Now, we'll focus on two key techniques: serial dilution and spectrophotometry. These methods are critical for the precision in the lab! By carefully diluting solutions and using spectrophotometric analysis, scientists can precisely determine concentrations. This information is vital for various applications, from pharmaceutical development to environmental monitoring. As we explore these methods further, we uncover their importance in scientific research and experimental analysis.



Well, let's start with dilution. The purpose of **dilution** is pretty straightforward: it's all about systematically reducing a solution's concentration to just the right level. This is essential because many reagents can be too high in concentrations, which can lead to inaccuracies or adverse effects if used directly. Imagine trying to measure out medication in a hospital without diluting it properly – it could lead to serious overdoses and harm to patients. That's why dilution is our trusty ally in ensuring accuracy and safety in various applications, from pharmaceutical development to environmental monitoring. Now, let's talk about **serial dilution**. This technique involves stepwise dilution to create a series of decreasing concentrations, each one reducing the concentration of the solution even further. When we visualize serial dilution, we see this beautiful gradient forming – kind of like watching colors fade away gradually, just like in Figure 9.1. This process isn't just for show, though; it's how we create solutions with pinpoint precision for all sorts of experiments and analyses in the lab.

Let's break down the steps of serial dilution using a practical example:

1. Begin with a known volume of a concentrated dye solution, such as 1 mL, in a labeled test tube.
2. Add 9 mL of solvent (typically water) to the original solution and thoroughly mix. This results in a 1:10 dilution, meaning the concentration is one-tenth of the original.



3. Take 1 mL from the 1:10 dilution and add it to another 9 mL of solvent. Now you have a 1:100 dilution.
4. Repeat this process, diluting the solution by a factor of 10 each time.
5. After multiple dilutions, you'll have a series of test tubes with decreasing concentrations, forming a concentration gradient from the original to highly diluted solutions.

This stepwise dilution process allows scientists to create solutions with precise concentrations suitable for various experiments, ensuring accurate and reliable results in analytical chemistry.

One way we use serial dilution is to compare an unknown solution to a concentration gradient. For example, let's consider a solution of red dye. By observing where it falls within the color gradient, we can estimate its concentration. This method isn't always precise, though, so we often need more accurate analytical techniques. Say you're experimenting to determine the concentration of a sugar solution. You prepare a series of sugar solutions with known concentrations using serial dilution. Then, you measure the absorbance of each solution using a spectrophotometer. By plotting the absorbance against concentration, you create a standard curve. When you measure the absorbance of an unknown sugar solution and compare it to the standard curve, you can accurately determine its concentration. Now, let's put this into context with a real-lab scenario. Imagine a researcher needs to quantify the concentration of an enzyme or molecule in an assay. By using serial dilutions, they can create a range of concentrations. Then they can compare their unknown enzyme concentration to this gradient. This approach is both cost-effective and efficient, minimizing waste while ensuring precise measurements.

Spectrophotometry is a powerful analytical technique used to quantify substances in solutions. **Spectrophotometers** operate on the principles of **transmission** and **absorbance**. They measure the amount of light that passes through a sample and the extent to which it absorbs light, respectively. **Absorbance values**, typically represented as peaks at specific wavelengths, directly correlate with the concentration of the substance being analyzed. Now, let's break down how spectrophotometry works. Spectrophotometers emit light of a specific wavelength through a sample solution. A **photodetector** then detects the amount of light absorbed by the solution. The instrument measures the absorbance of the solution, which is the logarithm of the ratio of incident light intensity to transmitted light intensity. In essence, spectrophotometry is at the core of quantitative analysis, allowing scientists to elucidate the presence and concentration of dissolved chemicals in solutions.

Consider an experiment where you're analyzing the concentration of a protein solution. You prepare a series of protein solutions with known concentrations and measure their absorbance at a specific wavelength using a spectrophotometer. The higher the protein concentration, the more light is absorbed, resulting in a higher absorbance value. By plotting absorbance against concentration for the standard solutions, you create a **standard curve**. Then, when you measure the absorbance of an unknown protein solution, you can accurately determine its concentration using the standard curve.

Next, we have to discuss the role of the **blank solution**. This **blank**, typically made of the solvent without any analyte, helps us account for any interference from the solvent or other components. By measuring the absorbance of the blank, we ensure accurate measurements of the target substance without any unwanted influences.

Now, how do you interpret unknown concentrations? To determine the concentration of an unknown solution, we use serial dilution to create a standard curve. This involves preparing a series of solutions with known concentrations and measuring their absorbance. By establishing a linear relationship between concentration and absorbance, we create a reference standard curve. Then, we compare the absorbance of the unknown solution to this curve, allowing us to precisely determine its concentration.

In summary, serial dilution and spectrophotometry are fundamental tools in biotechnology, enabling precise quantification of substances in solutions. Whether in pharmaceutical assays or environmental monitoring, these techniques empower scientists to uncover concentration mysteries with unparalleled accuracy.

### LAB ACTIVITY - PART 1 - Serial Dilution

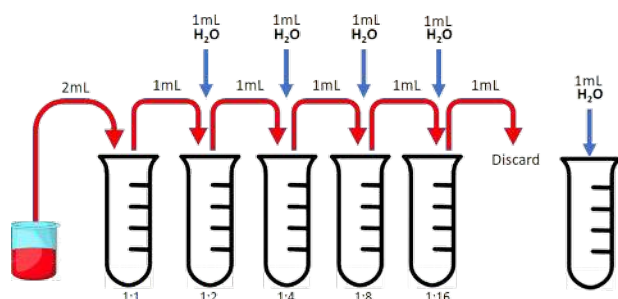
#### Materials

- One Cuvette holder (Grainger, 12C181)
- Six Uvettes (Fisher Scientific, E4099100008)
- 500mL of 0.5M of Copper (II) Sulfate (Flinn Scientific, C0108)
- One p1000 Micropipette.
- One box of p1000 Micropipette tips

#### Methods for 2-Fold Serial Dilution:

Today we will perform serial dilutions using copper (II) sulfate solution. This process is vital for various applications in scientific research, including accurately measuring the concentrations of substances in solution.

1. Take six cuvettes and label them from 1 through 6.
2. In cuvette 1, we'll add 2 milliliters (mL) of the 0.5M copper (II) sulfate solution.
3. For cuvettes 2 through 6, we'll add 1 mL of water to each.
4. Now, we'll start diluting our solution:
  - Take 1 mL of the solution from cuvette 1 and transfer it to cuvette 2. Repeat this transfer from cuvette 2 to 3, 3 to 4, and 4 to 5.
  - We won't transfer anything to cuvette 6. This cuvette serves as our blank.
5. Check if each cuvette has a final volume of 1 mL to maintain consistency.



## LAB ACTIVITY - PART 2 - Spectrophotometry

### Materials

- One Spectrophotometry
- Samples in Uvettes
- One Cuvette holder (Grainger, 12C181)
- One p1000 Micropipette.
- One box of p1000 Micropipette tips

### Spectrophotometry:

Now, let's move on to the next step of our experiment, where we'll utilize spectrophotometry. This tool allows us to measure the amount of light absorbed by our samples, giving us valuable insights into their properties.

1. Set the spectrophotometer to read at a wavelength of 662 nanometers (nm). This specific wavelength is the optimal absorbance for measuring Copper (II) Sulfate.
2. Examine the spectrophotometer and identify where the light source.
3. Check each cuvette to ensure that the clear side faces the light source.
4. Start by placing cuvette 6, our blank, into the cuvette reader.
5. Blank the spectrophotometer using cuvette 6 as our reference point.
6. Next, measure the absorbance of cuvettes 1 through 5, one by one.
7. As you measure each cuvette, carefully record the absorbance values.

| Wavelength (662nm) |            |
|--------------------|------------|
| Concentration (M)  | Absorbance |
| 0.5                |            |
| 0.25               |            |
| 0.125              |            |
| 0.0625             |            |
| 0.03125            |            |

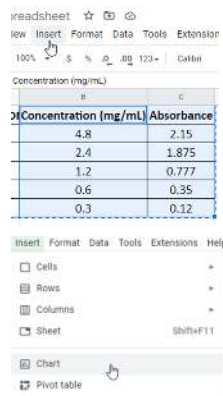
## LAB ACTIVITY - PART 3 - Standard Curve

### Standard Curve:

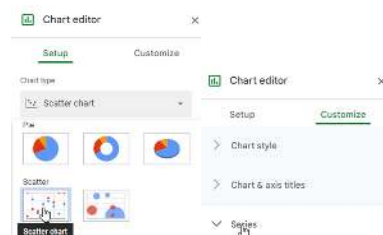
Now that we've completed the serial dilution and obtained absorbance values using spectrophotometry, it's time to take our analysis to create a standard curve. This curve will be generated using Excel spreadsheets.

| Concentration (M) | Absorbance (662nm) |
|-------------------|--------------------|
| 4.8               | 2.15               |
| 2.4               | 1.875              |
| 1.2               | 0.777              |
| 0.6               | 0.35               |
| 0.3               | 0.12               |

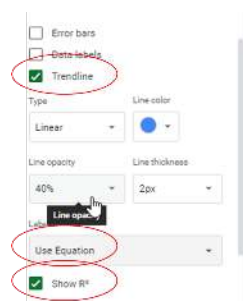
- Input your data, including the concentration values and their corresponding absorbance values, into a Google Sheet. It's crucial to include the label above these values with "Concentration (M)" and "Absorption (662nm)" to avoid graphing complications.



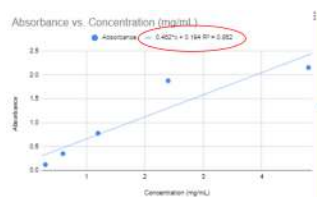
- Highlight the entire dataset, including both values and labels.
- Next, navigate to the "Insert" dropdown menu and select "Chart." This will bring up both the chart and the chart editor options.



- In the chart editor, under "Setup", ensure the chart type is set to "scatter chart".
- Moving to the "Customize" section, locate the "Series" option. Here, we'll select "Trendline" to add a trendline to our graph.
- Scroll down to the "Label" section and check the box to show the R2 value. This value indicates how well our data fits the trendline, serving as a measure of the reliability of our results.

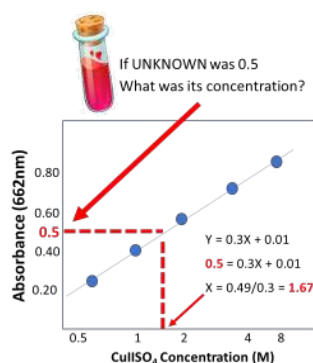


- Switch the option under "Label" to "Use Equation." This will display the equation of the trendline, which follows the form  $y = mx + b$ . This equation will help us determine the concentration of unknown samples.
- See the graph:



## LAB ACTIVITY - PART 3 - Determining Concentration

### Determining the Unknown Concentration:



Now that we've established our chart with the equation and R2 value, let's dive into interpreting our data and determining unknown concentrations.

- First, let's observe the R2 value. This value indicates how well our data fits the trendline. The closer it is to 1, the more reliable our results are. Typically, we trust the graph if the R2 value is above 0.95. If it falls below this threshold, it's recommended to consider redoing the serial dilution and make changes to your pipetting method.
- Next, let's apply the Equation. Let's say our example equation is  $y = 0.3x + 0.1$ , where  $y$  represents absorbance and  $x$  represents concentration. Suppose we measured an unknown Copper (II) sulfate solution via spectrophotometry and obtained an absorbance value of 0.5. Since absorbance is graphed on the Y-axis and concentration is graphed on the X-axis, we'll plug in 0.5 for  $Y$  in our equation:  $0.5 = 0.3x + 0.1$ .
- Solving for  $x$ , we find that  $x = 1.67$  M. This is our unknown concentration. We can apply this same equation to determine the concentrations of other unknown samples, providing valuable insights into their composition.

## LAB ACTIVITY - PART 4 - Dilution Maths

Alright, let's dive into dilution math and understand how to calculate the volumes of stock solution and water needed to achieve the desired concentration. Imagine you have purchased a therapeutic solution with a stock concentration of 100 grams per milliliter (g/mL). However, through previous data analysis, you've determined that the most effective therapeutic concentration is 1 g/mL. Now, the question is: How can you create more solutions at this optimal concentration when you only have the stock solution at 100 g/mL available?

First, let's discuss the formula we'll be using:  $C_1V_1 = C_2V_2$ , where  $C_1$  and  $C_2$  are the initial and final concentrations, respectively, and  $V_1$  and  $V_2$  are the initial and final volumes, respectively. Now, let's say we purchased 10 mL of a therapeutic stock solution with a concentration of 100g/mL, and we want to dilute it to a final concentration of 1g/mL. Our initial volume  $V_1$  is 10mL, and our final volume  $V_2$  is what we're trying to find. So, let's plug the numbers into our formula:  $C_1V_1 = C_2V_2$

$$(100\text{g/mL}) \times (10\text{mL}) = (1\text{g/mL}) \times V_2$$

Now, we can solve for  $V_2$ , which should be 1000mL total volume. So, we need to add approximately 990mL mL of water to dilute our 10 mL of stock solution to achieve a final volume of 1000mL and the final concentration of 1g/mL.

Now, let's move on to Example 2: Let's say we have a therapeutic stock solution with a concentration of 100g/mL, but we only need a volume of 200mL with a concentration of 1g/mL. We'll use the same formula,  $C_1V_1 = C_2V_2$ , but we need to find  $V_1$  this time.  $V_1$  is the volume of the stock solution we'll use. So, let's plug the numbers into our formula:  $C_1V_1 = C_2V_2$

$$(100\text{g/mL}) \times V_1 = (1\text{g/mL}) \times 200\text{mL}$$

Now, we can solve for  $V_1$ , which should be 2mL total volume. So, we need to use 2 mL of the stock solution and add 198 mL of water to achieve a final volume of 200 mL with a concentration of 1g/mL.

It's important to understand why we might choose bulk dilution or aliquots. In bulk dilution, like in Example 1, we dilute the entire stock solution to save time and convenience. However, in aliquot dilution, like in Example 2, we dilute smaller portions to avoid denaturation. This is because as some molecules become more diluted, they can become less stable, so diluting smaller portions helps maintain stability and long-term storage of our chemical stocks and reagents.

Go ahead and practice answering the problem sets below with an activity:

#### Materials

- Obtain three 50-mL beakers and fill it with 5mL of 0.5M CuIISO4
- Obtain one 50-mL beaker and fill it with 50mL of 0.5M CuIISO4
- Obtain one larger beaker and fill it with 500mL of water.

**Practice Problem #1:** How much **water** must you **add** to the stock to obtain a new final concentration?

#### STOCK

- You are given three beakers containing 5mL of 0.5M CuIISO4.

#### Final product

You want to create three different concentrations from these stocks:

- 50mM CuIISO4
- 100mM CuIISO4
- 250mM CuIISO4

**Practice Problem #2:** How much **stock and water** must you **mix** to obtain a new final concentration?

#### STOCK

- You are given one beaker containing 50mL of 0.5M CuIISO4.

#### Final product

You want to create three different concentrations with a total volume of 50mL from the stock solution:

- 50mL of 50mM CuIISO4
- 50mL of 100mM CuIISO4
- 50mL of 250mM CuIISO4

---

### LAB ACTIVITY - PART 5 - Application of Serial Dilution

Today we're going to perform serial dilutions using copper (II) sulfate solution. We'll start with a 2-fold dilution in six cuvettes, followed by a linear dilution in ten cuvettes.

#### Materials

- Microcentrifuge tubes
- One Cuvette holder (Grainger, 12C181)
- Six Uvettes (Fisher Scientific, E4099100008)
- 500mL of 0.5M of Copper (II) Sulfate (Flinn Scientific, C0108)
- One p1000 Micropipette.
- One box of p1000 Micropipette tips

#### Procedures:

- Get together with your group and decide on the diluted concentration of Copper (II) Sulfate. It should be between 100mM to 250mM. We'll need just 1mL of the solution.
- Write down the concentration you've chosen

- Dilute your solution and then measure its absorbance in the box below:

| Chosen Concentration (M) | Expected Absorbance |
|--------------------------|---------------------|
|                          |                     |

For the remainder steps, you will conduct a **2-fold serial dilution**

- We'll begin with a 0.5M copper (II) sulfate solution and dilute it in deionized water. Each subsequent cuvette will be diluted two-fold from the previous one.
- Label seven (7) cuvettes from 1 through 7.
  - Add 2mL of 0.5M copper (II) sulfate solution to cuvette 1.
  - Add 1mL of water to cuvettes 2 through 7.
  - Transfer 1mL of solution from cuvette 1 to cuvette 2, then from 2 to 3, 3 to 4, 4 to 5, and 5 to 6.
  - Discard the 1mL from cuvette 6.
  - Do not transfer any solution to cuvette 7, as it serves as a blank.
- After completing the dilution series, we'll analyze the resulting solutions using spectrophotometry, reading at 662nm, to generate graphs. Record your data using the table below:

**Standard Curve for Cu(II)SO<sub>4</sub>**

| Concentration (M) | Absorbance (Choose measurements between 600nm - 700nm) |
|-------------------|--------------------------------------------------------|
| 0.500             |                                                        |
| 0.250             |                                                        |
| 0.125             |                                                        |
| 0.063             |                                                        |
| 0.031             |                                                        |
| 0.016             |                                                        |

- We'll calculate the R-squared (R<sup>2</sup>) values to assess the linearity of our dilution series. Finally, we'll plug in the "**Expected Absorbance**" value into the  $y=mx+b$  equation from our graphs to determine the concentration by solving for x.
- We'll take the observed concentration value from the graph and compare that to your **Chosen Concentration** from the above table and solve to percent error:
  - Percent Error =  $(| \text{Chosen Concentration} - \text{observed Concentration} | / \text{Chosen Concentration}) \times 100\%$
- Was your calculated concentration similar to your **Chosen Concentration** of Copper (II) Sulfate?

### LAB ACTIVITY - PART 6 - Skill Evaluation 2 (20pts)

The instructor will create a **diluted UNKNOWN solution**, ranging between 100mM to 250mM

For this skill evaluation, you will conduct a **2-fold serial dilution**.

You will record two separate independent duplications for this run as only the best result will be input into your grade

- We'll begin with a 0.5M copper (II) sulfate solution and dilute it in deionized water. Each subsequent cuvette will be diluted two-fold from the previous one.
- After completing the dilution series, we'll analyze the resulting solutions using spectrophotometry, reading at 662nm, to generate graphs.
- We'll calculate the R-squared (R<sup>2</sup>) values to assess the linearity of our dilution series.
- Finally, we'll measure the absorbance of the **diluted UNKNOWN solution** from the instructor.
- We'll use the equation from our standard curve graphs to determine the concentration of the **diluted UNKNOWN solution**.
- We'll take the observed calculated concentration value and report that to the instructor

- The instructor will compare that to the expected value and solve to percent error:
  - $\text{Percent Error} = ((\text{Expected Value} - \text{Observed Value}) / \text{Expected Value}) \times 100\%$

**Grading rubric:**

- The instructor will grade you based on your percent error. Only the lowest percent error from your two duplicate results will be part of your grade
    - For example, if you obtained 8% error on one experiment and 5% error on another experiment, the instructor will only record the 5% error as part of your grade
    - Grade calculation: if your percent error was 5%, the final grade will be calculated as:  $20 - (5) = 15$  points.
  - A bonus of 2 points of extra credit will be given to any individual with a percent error under 1%.
    - If both duplicates were under 1%, then you will be granted 4 additional points to your final grade.
- 

**✓ Post-Lab Questions**

1. What challenges did you encounter when performing the dilutions? How did you overcome them?
2. How does understanding dilution and spectrophotometry apply to real-world laboratory settings?
3. What improvements could be made to increase the accuracy of your experiment

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## 10: Enzymes and pH Buffer

### Learning Objectives

- Students should understand the fundamental concepts of enzyme function, pH, and buffer systems. The goal is to prepare students for hands-on experimentation by reinforcing theoretical knowledge and ensuring they can make informed observations.

### Pre-Lab Questions

- What are enzymes, and how do they facilitate biochemical reactions?
- Explain the role of activation energy in enzymatic reactions.
- What is equilibrium in the context of enzyme activity, and how do enzymes affect reaction rates?
- Define pH and explain why pH stability is important for enzyme function.
- What are buffers, and how do they help maintain pH stability in biological systems?
- How does the Henderson-Hasselbalch equation help determine buffer capacity and pH adjustments?

### Note

- Review the concept of enzyme-substrate interactions.
- Read about the effects of pH on enzyme activity and how buffers stabilize pH.
- Familiarize yourself with the procedure for calibrating and using a pH meter.
- Review the Henderson-Hasselbalch equation and perform sample calculations for buffer preparation.
- Predict how changes in pH might affect enzyme activity and buffer efficiency.

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## Chapter Ten

### Enzymes and pH Buffer

In this Chapter, we're embarking on **enzymes**, those remarkable molecular machines that drive the biochemical processes essential for life. But before we delve into the details of how enzymes work, let's first understand the fundamental concept of substrate conversion into products. Imagine you're in a classroom, attempting to throw paper balls into a trash can. Now, think about the chances of success if you randomly toss the paper balls without aiming properly. Not very high, right? Well, in biochemical reactions, substrates must align correctly for reactions to occur effectively. Enzymes come to the rescue by guiding substrates into the right positions, much like someone helping you aim those paper balls directly into the trash can, increasing the likelihood of successful conversions.

However, it's crucial to understand that enzymes don't change the final outcome of reactions. Instead, they accelerate the reaction to **equilibrium**, which represents the peak point of substrate and product conversion. To illustrate this concept, envision tossing a bunch of papers into a trash can. However, if the trash is already full, no matter how many papers you attempt to toss in, they will simply fall off. This scenario mirrors the concept of equilibrium in enzymatic reactions, where substrate and product conversion reaches its peak. Furthermore, the reduction of transition energy can be likened to cycling from point A to point B. Imagine you're riding a bike from point A to point B. Without an enzyme, it's akin to pedaling uphill – a strenuous task. However, with the assistance of an enzyme, it's like cycling on a flat road – significantly easier. Enzymes achieve this by reducing the activation energy needed for reactions to proceed, facilitating the transition to equilibrium. Nonetheless, it's important to note that enzymes do not alter the equilibrium state itself.

Now, let's transition to our next topic: pH and its critical role in enzyme function. Enzymes, like many other molecules and proteins, are sensitive to changes in pH. Most life forms thrive within a relatively narrow pH range, and extreme pH levels can lead to protein denaturation and precipitation. This highlights the importance of maintaining pH balance within living systems. But how do living organisms maintain pH balance, especially in the face of dramatic pH changes? **Buffers** – nature's pH regulators. Buffers are solutions capable of resisting changes in pH by absorbing or releasing hydrogen ions ( $H^+$ ) or hydroxide ions ( $OH^-$ ).

To understand buffers better, let's compare strong acids like hydrochloric acid (HCl) to weak acids like acetic acid. When you add HCl to water, it fully dissociates, rapidly lowering the pH. In contrast, weak acids like acetic acid only partially dissociate, allowing



them to maintain pH stability by absorbing excess  $H^+$  ions. This buffering capacity is essential for preventing drastic changes in pH, akin to a trashcan absorbing additional trash to keep the environment clean.

However, it's crucial to recognize that even buffers have their limits. Every weak acid or base has a specific **buffer zone**, akin to holding a trash can steady. Within this buffer zone, the molecule can effectively maintain pH stability, much like keeping the trash can balanced without spilling its contents. However, when you go beyond this buffer zone, it's like piling too much trash onto the can. Eventually, the can tilts, and all the trash spills out. Similarly, when adjusting the pH from 7 to 6, adding slightly more acid can cause the pH to plummet to 2 – a clear indication of surpassing the buffer zone's capacity to stabilize pH.

Now, let's apply our understanding of pH and buffers to the laboratory setting. Suppose we want to manipulate pH to create optimal conditions for enzymatic reactions. In this scenario, we can use the **Henderson-Hasselbalch equation** to calculate the precise amount of acid needed to achieve a target pH. For example, let's say we mixed 1.67mL of 1M acetate with 1mL of 1M acetic acid, with a  $pK_a$  value of 4.76. By plugging these values into the Henderson-Hasselbalch equation, we can determine the resulting pH of the solution.

**Step 1:** Calculate the moles of acetate (buffer):

$$\text{Buffer} = \text{Volume} \times \text{Concentration}$$

$$= 0.00167 \text{ L} \times 1 \text{ mol/L}$$

$$= 0.00167 \text{ mol}$$

**Step 2:** Calculate the moles of acetic acid (weak acid):

$$\text{Acid} = \text{Volume} \times \text{Concentration}$$

$$= 0.001 \text{ L} \times 1 \text{ mol/L}$$

$$= 0.001 \text{ mol}$$

**Step 3:** Substitute the values into the Henderson-Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log([A^-]/[HA])$$

$$\text{pH} = 4.76 + \log(0.00167/0.001)$$

**Step 4:** Perform the logarithmic calculation:

$$\text{pH} = 4.76 + \log(1.67)$$

$$\text{pH} \approx 4.76 + 0.2228$$

$$\text{pH} \approx 4.983$$

But what if we want to adjust the pH to a specific value, such as pH 5? How much volume of **1M acetic acid** should you add to mix with **1.67mL of 1M acetate** to obtain pH 5? By rearranging the Henderson-Hasselbalch equation and solving for the acid concentration, we can calculate the volume of acid needed to achieve the desired pH. In this case, we find that only 1.05mL of acid is required, demonstrating the precision and utility of buffer calculations in laboratory experiments.

**Step 1:** Substitute the values into the Henderson-Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log([A^-]/[HA])$$

$$5 = 4.76 + \log[(0.00167\text{L})(1\text{mol/L})/(1\text{mol/L})(A-L)]$$

**Step 2:** Rearrange the equation to solve for the concentration of acetic acid (HA):

$$5 - 4.76 = \log(0.00167/A-L)$$

$$0.24 = \log(0.00167/A-L)$$

**Step 3:** Rewrite the equation in exponential form:

$$100.24 = 0.00167/A-L$$

**Step 4:** Solve for the concentration of acetic acid (HA):

$$A-L = 0.00167/100.24$$

$$\approx 0.00167/1.738$$

$$A-L \approx 0.00096$$

**Step 5:** Divide 0.00096M by the Molarity of acid to determine the volume of acetic acid needed:

$$A-L \approx 0.00096 \approx 0.00096 \text{ L or } 0.96\text{mL}$$

In conclusion, enzymes are the unsung heroes of biochemical reactions, driving essential processes with finesse and precision. Yet, their activity is intricately linked to factors such as pH, which can profoundly influence enzymatic function. Buffers play a crucial role in maintaining pH stability, ensuring optimal conditions for enzyme activity. By understanding the interplay between enzymes, pH, and buffers, we unlock the secrets of life's biochemical complexity and pave the way for groundbreaking discoveries in science and medicine.

### LAB ACTIVITY - PART 1 - pH meter Calibration and Verification

Why it's crucial to calibrate your pH meter and verify it with known standards? **pH meters** measure the acidity or alkalinity of a solution based on the voltage generated by a pH-sensitive electrode. Over time, factors such as electrode wear, contamination, or drift can affect the accuracy of readings. **Calibration** ensures that the pH meter provides accurate and reliable measurements. **Verifying** the pH meter with known buffer standards validates its accuracy and reliability. Known buffer solutions with precise pH values serve as reference points for verifying the performance of the pH meter. Verification with known standards is essential for quality control purposes, especially in laboratory settings. It ensures that the pH meter meets quality assurance standards and adheres to regulatory requirements for accurate pH measurements.

#### Instruction: Preparation Before Using the pH meter:

1. Turn on the pH meter and allow it to warm up.
2. Rinse the pH electrode with distilled water to remove any residue or contaminants.
3. Immerse the electrode in a storage solution or distilled water to hydrate the sensitive electrode until you are ready to use it.

#### Instruction: Using the Probe:

1. Gently wipe off excess liquid, but only dap to dry the sensitive electrode.
2. Carefully submerge the electrode into the sample solution.
3. Allow the pH reading to stabilize, and write down the pH value.
  1. If it's not stabilizing within 1 minute, go ahead and write down what you see.

#### Instruction: Cleaning After Use:

1. After measurements are complete, remove the pH electrode from the sample solution.
2. Rinse the electrode with distilled water to remove any residual sample solution.
3. Gently wipe off excess liquid, but only dap to dry the sensitive electrode.
4. Store the pH electrode in a storage solution or distilled water until the next use.

#### Instructions: Calibrate Your pH Meter:

1. Prepare buffer solutions with pH values of 4.01, 7.00, and 10.01.
2. Immerse the pH electrode of the meter into each buffer solution sequentially.
3. Follow the manufacturer's instructions to calibrate the pH meter using the provided calibration controls.
4. Ensure that the meter is properly calibrated and stable before proceeding to the next step.

#### Instruction: Verify with Buffer Solutions (AFTER CALIBRATION IS COMPLETED)

1. Re-measure the same buffer solution (4.01, 7.00, and 10.01), **without calibration**
2. Immerse the pH electrode into the buffer solution with a pH of 4.01.
3. Record the pH reading displayed on the meter.
4. Repeat the process with buffer solutions of pH 7.00 and 10.01, recording the pH readings each time.

## 5. Determine Absolute Error:

1. Absolute Error = |Measured pH - Known pH|
2. Tape the absolute error data on the pH meter.

### Summary instruction:

- Calibrate your pH meter.
  - Standardize with buffer 4.01, 7.00, and 10.01
- Determine the **absolute error** for your pH meter.
  - Check with pH 4.01
  - Check with pH 7.00
  - Check with pH 10.01
- Tag the pH machine with its **absolute error**

Note: You do not need to calibrate again if you believe it is consistent with the standards

$$(\text{Absolute error}) = [(\text{Theoretical pH}) - (\text{Actual pH})]$$

## LAB ACTIVITY - PART 2 - pH Calculation and measurement

### Materials:

- 1M Acetic acid
- 1M Sodium Acetate

### Theoretical Calculation:

1. Calculate and determine the quantities of 0.1M acetic acid needed to achieve the theoretical pH values of 4.76, 6, 7, and 8 given a volume of 10mL 0.1M Sodium acetate.
  - Utilize the Henderson-Hasselbalch equation.

### Actual Experiment:

1. Dilute the 1M Acetic acid and 1M Sodium Acetate
2. Mix in the calculated amounts of 0.1M acetic acid with 10 mL of 0.1M Sodium acetate in a small beaker for each target pH value (4.76, 5, 5.5, and 6)
3. Use a magnetic stir to mix to achieve a homogeneous solution.
4. Use a calibrated pH meter to measure the pH of the mixed solution for each solution.
5. Record the measured pH values obtained.

| Theoretical pH | 0.1M Sodium acetate (mL) | 0.1M acetic acid (mL) | Actual pH |
|----------------|--------------------------|-----------------------|-----------|
| 4.76           | 10                       |                       |           |
| 6              | 10                       |                       |           |
| 7              | 10                       |                       |           |
| 8              | 10                       |                       |           |

## LAB ACTIVITY - PART 3 - Skill Evaluation 3 (20 points)

### Materials:

- 1M Acetic acid
- 1M Sodium Acetate

### Calculation:

- Calculate and record the exact volumes of 0.1M acetic acid required to achieve pH values of 5.8 given a volume of 20mL of 0.1M Sodium acetate.

**Testing:**

1. Dilute 1M Sodium acetate and 1M acetic acid solutions to 0.1M concentrations.
2. Measure out and mix in the exact volumes of the 0.1M Sodium acetate and 0.1M acetic acid solutions determined from the calculations to obtain a pH value of 5.8.

**Grading:**

1. The instructor will measure and record your pH.
  2. Deduct the percent error from the total score as a grading criterion.
    - If the percent error is 5%, then the total score will be 15/20.
- 
- 

**LAB ACTIVITY - PART 4 - Practicum 1 (60 points)****Dilution Activity (20 points):**

- **Materials:**
    - Bottle of 0.5M stock CuISO<sub>4</sub> solution
    - Unknown CuISO<sub>4</sub> solution was created by the instructor (100mM to 250mM).
    - Cuvette and spectrophotometer
  - **Goal:** Determine the concentration of the unknown solution
    - Create a 2-Fold serial dilution
    - Measure your dilution and the unknown via Spectrophotometry.
    - Create a standard curve graph. Calculate the concentration of your unknown.
  - **Evaluation:** The percent error from your calculated concentration and the actual concentration will be deducted from the total score.
    - If the percent error is 5%, then the total score will be 15/20.
- 

**Dilution Re-creation (20 points):**

- **Materials:**
  - Bottle of 0.5M stock CuISO<sub>4</sub> solution
  - Distilled water.
  - Cuvette and spectrophotometer
- **Goal:** Recreate the unknown solution.
  - Apply  $C_1V_1 = C_2V_2$  to calculate the volume of 0.5M CuISO<sub>4</sub> and water to mix to obtain the predicted unknown concentration from Part 1.
  - Dilute the 0.5M CuISO<sub>4</sub> with water to have the same concentration as Part 1.
    - Make sure the total volume is 1 mL in a cuvette.
  - Submit the cuvette to your instructor.
- **Evaluation:** The percent error from your recreated absorbance and your previous absorbance will be deducted from the total score.
  - If the percent error is 5%, then the total score will be 15/20.

**Summary instruction:**

| Standard Curve    |                    |
|-------------------|--------------------|
| Concentration (M) | Absorbance (620nm) |
| 0.5               |                    |
| 0.25              |                    |
| 0.125             |                    |
| 0.0625            |                    |
| 0.03125           |                    |

#### Dilution Activity (20 points):

- **Goal:** Determine the **concentration** of the unknown solution
  - Create a 2-Fold serial dilution
  - Measure your dilution and the unknown via Spectrophotometry.
  - Create a standard curve graph. Calculate the concentration of your unknown.

#### Dilution Re-creation (20 points):

- **Goal:** Recreate the unknown solution.
  - Calculate the volume of 0.5M CuI<sub>2</sub>SO<sub>4</sub> and water to **mix** together
  - Make sure the total volume is 1 mL in a cuvette.
  - Submit the cuvette to your instructor.

| Unknown            |                   | Instructor only       |                              |                       |
|--------------------|-------------------|-----------------------|------------------------------|-----------------------|
| Absorbance (620nm) | Concentration (M) | Concentration % error | Recreated Absorbance (620nm) | Recreated Abs % error |
|                    |                   |                       |                              |                       |

### pH Activity (20 points): **pH meter is not allowed during this examination**

- **Materials:**
  - Bottle of 0.5M Sodium acetate
  - Bottle of 0.5M acetic acid.
- **Goal:** Obtain a pH of 5.
  - Obtain a 50mL beaker or Conical tube, and add 10mL of 0.5M Sodium acetate
  - Calculate the theoretical volume of 0.5M acetic acid required to obtain a pH 5 to be mixed in with the 10 mL of 0.5M Sodium acetate.
  - Mix the calculated volume and submit the buffer to your instructor.
- **Evaluation:** The percent error from your pH will be deducted from the total score.
- If the percent error is 5%, then the total score will be 15/20.

| pH | Student's pH measurements | Percent error (%) |
|----|---------------------------|-------------------|
| 5  |                           |                   |

### ? Post-Lab Questions

- Summarize the key observations you made during the experiment. How did enzyme activity change with different pH levels?
- How did calibration of the pH meter impact the accuracy of your measurements? What would happen if the meter were not calibrated properly?
- Explain the buffering capacity observed in your experiment. Did the buffer effectively resist changes in pH? Why or why not?
- Based on your results, how do extreme pH levels impact enzymatic function? What molecular interactions are responsible for these effects?
- Reflect on the importance of the Henderson-Hasselbalch equation in predicting and controlling pH in biological systems. How did your experimental results compare to your calculated predictions?

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## 11: Creating Buffer Solutions

### Learning Objectives

- Understand the concept and importance of buffer solutions in biochemical and biotechnological processes.
- Learn how to calculate the required amounts of components to prepare a buffer solution of a specific concentration and pH.
- Familiarize yourself with the procedures for preparing and adjusting buffer solutions.

### Note

- Review the principles of acid-base equilibria, focusing on weak acids and their conjugate bases.
- Study the Henderson-Hasselbalch equation and its application in buffer calculations.
- Read about the role of buffers in maintaining pH stability in various experimental contexts.

### Pre-Lab Questions

1. Explain how a buffer solution resists changes in pH upon the addition of small amounts of acid or base.
2. Why would a combination of a strong acid like HCl and a strong base like NaOH be unsuitable for creating a buffer solution?
3. Given a weak acid with a known pKa, calculate the ratio of conjugate base to acid required to prepare a buffer solution at a desired pH.
4. Describe the steps you would take to prepare 100 mL of a 0.1 M acetate buffer at pH 4.75 using acetic acid and sodium acetate.

---

## Chapter Eleven

### Creating Buffer Solutions

Today we're going to dive into the fascinating world of laboratory science. One of the key things we'll be exploring is the **creation of buffer solutions** with just the right concentrations. Why is this so important, you ask? Well, imagine you're conducting an experiment and your solution isn't at the exact concentration you need. It could throw off your results completely! That's where buffers come in. Buffers will help keep the pH levels stable, which is important for all sorts of experiments, especially those involving biochemical and biotechnological processes. So, buckle up as we go into the nitty-gritty of creating buffers. We'll go through each step, look at some real-life examples, and make sure you leave here feeling like a buffer expert ready to conquer any experiment!

Alright, let's talk about **buffer solutions**. A buffer solution is made by mixing a weak acid with its conjugate base or a weak base with its conjugate acid. This mixture is important because it can resist changes in pH when we add a little bit of acid or base to it. Now, why do we need buffers? Well, in the world of biochemical assays, enzymatic reactions, and cell culture studies, keeping the pH stable is key for things to work properly. Imagine you're trying to do an enzymatic reaction, and it needs to happen at a pH of 7.4. If the pH starts swinging all over the place, your reaction won't go as planned. That's where buffers come in handy. For example, let's say you're working with acetic acid and acetate ions. When you're doing your experiment and you accidentally add a little extra acid or base, the buffer will absorb those changes, keeping the pH steady at 7.4 and making sure your reaction runs smoothly.

Let's walk through the steps using a real-life example: preparing a 10 mL solution of 5 mM NaCl with a pH of 7.5. This process will give us a glimpse into the meticulous calculations and considerations involved in achieving the perfect concentration and pH.

#### Step 1: Determine Molecular Weight

- First, we need to figure out the **molecular weight** of our solute, which is NaCl. We know that sodium (Na) has a molecular weight of around 22.989 g/mol, and chlorine (Cl) weighs about 35.45 g/mol. So, when we add them together, we get a total molecular weight of NaCl, which is 58.439 g/mol.

#### Step 2: Convert Concentration Units

- Now that we have our molecular weight, let's convert our desired concentration from millimolar (mM) to **molar (M) units**. For a 5 mM NaCl solution, this equals 0.005 M (remember, 1 mM = 0.001 M).

### Step 3: Convert Volume Units

- Our solution volume is given as 10 mL, but we need to work with **liters (L)**. So, we'll divide 10 mL by 1000 to get 0.01 L.

### Step 4: Calculate Mass of Solute

- Now, let's put it all together to find out how much NaCl we need. We'll **multiply the molecular weight, molarity, and volume**:
  - Mass =  $58.439 \text{ g/mol} \times 0.005 \text{ mol/L} \times 0.01 \text{ L} = 0.0029 \text{ g}$

### Summary of calculation steps from Molarity:

- Step 1 – Find the **molecular weight (g/mol)**
  - Na – 22.989g/mol and Cl – 35.45g/mol
  - Total molecular weight = **58.439g/mol**
- Step 2 – Convert **mM** units to **mol/L**
  - 5mM → 0.005M → **0.005mol/L**
- Step 3 – Convert **mL** units to **L**
  - 10mL → **0.01L**
- Step 4 – Multiple step 1, 2 and 3 together
  - 58.439g/mol** x **0.005mol/L** x **0.01L** = **0.0029g**



Image of table salt from <https://unsplash.com/@tinkerman>

**Percentage concentration** is another way to talk about how strong our buffer solution is. For example, imagine we're making a 20% glycerol solution to use as a cryoprotectant for cell culture. To make 50 mL of this solution, we'd mix 10 mL of glycerol with 40 mL of water. This way, glycerol makes up 20% of the total volume.

Now, let's explore how we can express the concentration of our buffer solutions using percentages. Imagine we need to prepare a 10% NaCl solution with a pH of 7.5. Here's how:

### Step 1: Convert Percentage to Ratio:

- We start by turning the percentage into a decimal ratio. So, 10% becomes 0.10.

### Step 2: Calculate Mass of Solute:

- Next, we multiply the ratio by the volume to find out how much NaCl we need. For example, if we're making a 10 mL solution, we'd add 1 mL of NaCl ( $10 \times 0.1$ ). Since 1 mL of water is equivalent to 1 g, we add 1 g of NaCl to prepare the solution.

### Summary of calculation steps from percentage:

Step 1 – Convert percentage into ratio  
 $10\% / 100\% \rightarrow 0.10$   
 Step 2 – Multiply the ratio to the volume  
 $10\text{mL} \times 0.10 = 1\text{mL}$   
 Step 3 – Note that '1mL of water = 1g water'  
 $1\text{mL} \rightarrow 1\text{g}$  of salt

We only need to add **1g** of salt



Image of table salt from <https://unsplash.com/@tinkerman>

### Math Review Questions:

- How many grams of salt do I need to make: 10mL of 1M NaCl
- How many grams of salt do I need to make: 30mL of 0.5M NaCl
- How many grams of salt do I need to make: 10mL of 10% NaCl
- How many grams of salt do I need to make: 30mL of 50% NaCl

When we're making a buffer solution, we have to be mindful of **volume adjustments**. Adding more acid or base to adjust the pH can change the total volume of our solution, possibly diluting it beyond our desired concentration. So, it's smart to start with a

solution that's a bit below our target volume, usually about 80% full. Then, once we've adjusted the pH, we can add more solvent to reach our desired volume without changing the concentration. For example, let's say we're making a phosphate buffer with a pH of 7.4. After adding our calculated amounts of monosodium phosphate and disodium phosphate, we can then begin to add acid or basic solution to reach the right pH. However, we then noticed the volume had gone up a bit due to the acid or base we added during mixing. By leaving some room for volume adjustments at the start, we can handle these changes without messing up our final concentration.

Let's focus on a smart strategy for preparing buffer solutions that can save us a lot of time and resources. Instead of making each buffer solution separately, we can use what's called a **stock solution**. Imagine we have a whole bunch of experiments coming up, and each one needs a buffer with a slightly different pH. Instead of mixing up each buffer from scratch every time, we can make one big batch of a higher concentration buffer called a **stock solution**. Once we have our stock solution, we can dilute smaller amounts of it to get the exact pH we need for each experiment. This way, we're not spending all our time mixing up buffer solutions over and over again. Plus, it ensures that all our experiments are consistent because we're using the same stock solution every time.

Let me give you an example: say we need four 40 mL solutions of a 20% salt solution. We have two options: (1) We could make each 40 mL solution separately, which would take a lot of time and effort. (2) Or, we could make one big batch of, let's say, 200 mL of the 20% salt solution. Then, we can divide it into four equal parts, called **aliquots**, each measuring 40 mL. This way, we're saving time and ensuring that all our solutions are the same. But wait, there's more! We can also make highly concentrated solutions and then dilute them later as needed. For example, if we need those same four 40 mL solutions of 20% salt solution, we could make a 40 mL solution of 80% salt instead. Then, when we're ready to use it, we can take a small amount of the 80% solution and mix it with water to dilute it down to the 20% concentration we need. So, remember, when it comes to preparing buffer solutions, using stock solutions and strategic dilution techniques can save us time, and effort, and ensure consistency across our experiments.

Finally, let's talk about some important practices when it comes to preparing buffer solutions and optimizing our lab workflows. In the lab, it's crucial to label and document our solutions correctly. A well-labeled solution should include key information like the chemical name, concentration, date of preparation, and who made it. This might seem like a small detail, but it's super important. Proper labeling ensures that everyone knows exactly what's in each solution, which helps prevent mistakes and makes our experiments more reproducible.

In conclusion, preparing buffer solutions is a careful process that requires attention to detail and strategic thinking. By following systematic procedures, labeling our solutions properly, and using smart preparation techniques, we can ensure the stability, efficacy, and reproducibility of our buffer solutions in all sorts of lab experiments. So, let's keep mastering the art of buffer preparation to advance our scientific research and innovation!

### LAB ACTIVITY - PART 1 - Stock Solution

Students will learn how to prepare a concentrated stock solution of copper(II) sulfate ( $\text{CuIISO}_4$ ), dilute it to a desired concentration, aliquot it into cuvettes, and measure its absorbance using spectrophotometry. Additionally, they will perform quality control by assessing the consistency of absorbance readings.

Materials:

- Copper(II) sulfate ( $\text{CuIISO}_4$ ) powder
- Distilled water
- Analytical Balance
- Microcentrifuge tubes and small beakers
- Cuvettes
- Spectrophotometer
- Pipettes and pipette tips
- Gloves and safety goggles

Instructions:

#### 1. Preparation of 10mL of 0.5M $\text{CuIISO}_4$ Concentrated Stock:

- Weigh out the appropriate amount of  $\text{CuIISO}_4$  powder using a balance
- Transfer the weighed  $\text{CuIISO}_4$  powder into a beaker.



- Add distilled water and mix thoroughly until the powder is completely dissolved.

## 2. Dilution to Prepare 10mL of 0.1M CuISO4 Solution:

- Prepare a 0.1M solution of CuISO4 by diluting 1 mL of the 0.5M stock solution with 9mL of distilled water.

## 3. Aliquoting and Measurement in Cuvettes:

- Pipette 1 mL of the 0.1M CuISO4 solution into a cuvette.
- Pipette 1mL of distilled water into another cuvette
- Wipe the exterior of the cuvettes to ensure no solution is present outside.
- Blank your water into the spectrophotometer
- Then record the absorbance for copper(II) sulfate (662nm).

## 4. Quality Assurance and Quality Control (QA/QC):

- Compare the Absorbance readings with your colleagues. Calculate the standard deviation between you and your colleagues.

## LAB ACTIVITY - PART 2 - Skill Evaluation 4 (20 points)

The objective of this skill evaluation is to assess your ability to accurately prepare copper(II) sulfate (CuISO4) stock solutions at various concentrations and perform spectrophotometric measurements to determine absorbance values.

### Instructions:

\*Students are only allowed up to 2 submissions of their examination.

1. Prepare 0.6M following CuISO4 stock solutions at a volume of 20 mL each:
2. From the stock solution, dilute into the following concentration and volume:
  - 1mL of 0.5M
  - 1mL of 0.4M
  - 1mL of 0.3M
  - 1mL of 0.2M
  - 1mL of 0.1M
3. Then record the absorbance for all your samples at 662nm
  - Pipette 1mL of distilled water into a cuvette to blank the spectrophotometer
  - Transfer each sample into a cuvette and measure the absorbance at 662nm
4. **Evaluation Criteria:** Points will be based on the R2 value created from the standard curve graph
  - Every 0.01 ratio off is -1pt. (i.e. R2 value of 0.95 will be 15/20.)

### Summary of the activity:

1. Create a **0.6M CuISO4 Stock** (Per group)
2. From your **stock**, create the following dilutions
  - 1mL of 0.5M
  - 1mL of 0.4M
  - 1mL of 0.3M
  - 1mL of 0.2M
  - 1mL of 0.1M
3. Measure each dilutions and the **stock** via spectrophotometry
4. **Grading:** Points are based on R<sup>2</sup> values.

| Concentration | Abs |
|---------------|-----|
| 0.1M          |     |
| 0.2M          |     |
| 0.3M          |     |
| 0.4M          |     |
| 0.5M          |     |
| 0.6M          |     |

## ? Post-Lab Questions

1. How did the experimentally determined pH of your buffer solutions compare to the theoretical pH values? Discuss any discrepancies.
2. What factors could have contributed to variations in absorbance readings among different groups?
3. Explain the importance of using a blank solution in spectrophotometric measurements.

4. Discuss the significance of the  $R^2$  value in your standard curve. What does it indicate about your data?
5. Reflect on the process of preparing buffer solutions. What challenges did you encounter, and how did you address them?
6. Consider the importance of proper labeling and documentation in the laboratory. How does this practice contribute to the reproducibility and reliability of experimental results?
7. Think about how the skills and knowledge gained from this lab can be applied to future experiments and real-world scenarios.

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## 12: Advance Solution Preparation

### Learning Objectives

- Understand the principles and procedures for preparing microbiological growth media.
- Learn the correct use of an autoclave for sterilization purposes.
- Comprehend the importance of pH in biological systems and the methods for adjusting and measuring pH levels.
- Gain proficiency in preparing buffer solutions and understanding their roles in maintaining pH stability.

### Note

- Review the theory behind autoclaving, including standard conditions and the importance of sterilization in microbiology.
- Study the concepts of acids, bases, and pH, focusing on their relevance in biological experiments.
- Explore the composition and function of buffer solutions in biochemical processes.

### Pre-Lab Questions

1. What are the standard conditions (temperature, pressure, and time) required for effective autoclaving?
2. Why is it crucial to autoclave media before using it for culturing microorganisms?
3. Define a buffer solution and explain how it maintains pH stability in a solution.
4. Calculate the amounts of a weak acid and its conjugate base needed to prepare a buffer solution with a specific pH.

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## Chapter Twelve

### Advance Solution Preparation

Let's discuss the importance of precision and accuracy in scientific research, particularly when it comes to preparing solutions for experiments and assays. When we talk about **advanced solution preparation**, we're referring to careful attention to detail, following standard operating procedures (SOPs) to a tee, and using specialized equipment to ensure that our results are consistent and trustworthy. Today, we're going to explore the step-by-step procedures (SOPs) for preparing some common laboratory solutions that you'll encounter in your scientific journey. These include media for growing microorganisms, buffers with carefully adjusted pH levels, and gels used in a technique called electrophoresis.

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### LAB ACTIVITY - PART 1 - Preparation with Autoclave

#### Autoclaving:

Let's talk about **autoclaving**, which is a fundamental step in preparing laboratory solutions. Autoclaving is crucial because it sterilizes our equipment, media, and solutions, getting rid of any pesky microbial contaminants that could mess up our experiments. How does it work? Well, we use an autoclave, which is like a pressure cooker or a steam machine. It heats water to produce high-pressure steam, and we subject our solutions to this steam at temperatures of about 121°C for a set amount of time, usually between 15 to 30 minutes. This high temperature and pressure combo effectively kills bacteria, fungi, and other microorganisms, ensuring that our lab materials are sterile and ready for use in our experiments.

Alright, let's walk through the process of preparing LB broth and agar plates for growing bacteria, and then how we autoclave them to ensure they're sterile and ready for our experiments.

The objective is to prepare 50mL of Luria-Bertani (LB) broth and 30mL of Luria-Bertani (LB) agar plates using autoclave sterilization.

#### Materials:

- 125mL Erlenmeyer Flask (Carolina, #721149A)
- Tryptone (Carolina, #216741)
- Yeast Extract (Carolina, #216746)
- NaCl (Flinn Scientific, #S0064)

- Aluminum foil (Carolina, #974844)
- LB Broth Agar Base (Carolina, 216701)
- Petri dish (Carolina, #741251)
- Autoclave

#### 50mL of Luria-Bertani (LB) Broth Preparation:

- First, we need to gather our materials, including a 125mL Erlenmeyer flask.
  - Each group gathers one 125mL Erlenmeyer Flask.
  - Ensure the availability of tryptone, yeast extract, NaCl, and aluminum foil.
- We'll add the ingredients for LB broth: 1% tryptone, 0.5% yeast extract, and 1% NaCl. These provide the nutrients bacteria need to grow.
  - Measure Ingredients:
    - Add 0.5g of tryptone (1% concentration) into the Erlenmeyer Flask.
    - Add 0.25g of yeast extract (0.5% concentration) into the same flask.
    - Add 0.5g of NaCl (1% concentration) into the flask.
- Gently **swirl** the flask to ensure a thorough mixing of the ingredients.
  - The solution won't dissolve completely until after you autoclave.
- To prevent any contamination, we'll cover the flask with aluminum foil.
- Now, it's time to sterilize our LB broth. We'll place the flask in the autoclave and set it to run at 121°C for 25 minutes.
  - Cap the flask tightly with aluminum foil to prevent contamination during autoclaving.
  - Place the capped flask in the autoclave, and autoclave for 25 minutes at 120°C.
- After autoclaving, we need to let the LB broth cool down to room temperature before we can use it to grow bacteria.

#### 30mL Luria-Bertani (LB) Agar Plate Preparation:

- Similarly, we'll prepare our LB agar solution in another 125mL Erlenmeyer flask.
  - Each group gathers one 125mL Erlenmeyer Flask.
  - Ensure the availability of LB Broth Agar Base and aluminum foil.
- This time, we'll add 3% Luria Broth Agar Base to the flask. Agar makes the solution solid, providing a surface for bacteria to grow on.
  - Add 0.9g of LB Broth Agar Base (3% concentration) into the Erlenmeyer Flask.
  - Add 30mL of distilled water to the flask.
  - **Swirl** the flask gently to dissolve the agar base.
    - The solution won't dissolve completely until after you autoclave.
- Just like before, we'll cover the flask with aluminum foil to keep it clean.
- The flask goes into the autoclave, again at 121°C for 25 minutes, to make sure it's sterile.
  - Cap the flask tightly with aluminum foil to prevent contamination during autoclaving.
  - Place the capped flask in the autoclave, and autoclave for 25 minutes at 120°C.
- After autoclaving, we need to cool the agar solution to around 42°C. This temperature is important because it's hot enough to keep the agar liquid, but not so hot that it damages any heat-sensitive components and molecules.
- Once cooled to touch, we pour the agar solution into sterile petri dishes, where it will solidify. These plates are now ready for use in our bacterial culture experiments.
  - Carefully pour the cooled agar solution into petri dishes.
  - Leave the plates undisturbed until the agar solidifies.

So, there you have it! With these steps, we can prepare LB broth and agar plates, autoclave them to ensure they're free of contaminants, and create the perfect environment for growing bacteria in our experiments.

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#### LAB ACTIVITY - PART 2 - Preparation with pH Adjustment

Next, let's dive into pH adjustment, an essential skill for any scientist working in the lab. Whether you're conducting experiments in biochemistry or microbiology, understanding how to fine-tune pH levels is crucial for maintaining the right conditions for reactions to occur. Why do we care about pH adjustment? Well, imagine trying to bake a cake without measuring the ingredients properly. Just like how the right amount of sugar and flour is essential for a perfect cake, the correct pH level is crucial for many experiments to work effectively. So, what exactly is **pH adjustment**? It's the process of adding either acids or bases to a solution to reach a specific pH level. Once we've added the acids or bases, we need to mix everything up thoroughly and then double-check our work using a pH meter or pH indicator.

The objective is to prepare solutions with specific pH levels using Tris buffer and Sodium Acetate/Acetic acid buffer solutions.

#### Materials:

- Tris (Flinn Scientific, #FB2393)
- 0.1M Sodium Acetate solution
- 0.1M Acetic acid solution
- Distilled water
- Serological pipettes
- Micropipettes
- Beakers
- pH meter
- Stirring rod or magnetic stirrer

#### Procedure: Preparation of Tris Buffers

Let's break it down with an example: Your goal is to create 0.5M Tris buffer solution at pH 4, pH 7, and pH 10. Tris buffers are super versatile and can maintain a stable pH across different temperatures, making them perfect for many biochemical and molecular biology experiments. Here's a step-by-step guide, or Standard Operating Procedure (SOP), for preparing Tris buffers:

- **Step 1:** Gather your materials and prepare a 20mL of 0.5M Tris base solution.
  - Using a digital balance, weigh out the appropriate amount of Tris into a clean beaker
    - 1.21g of Tris base
  - Add distilled water to the Tris base, bringing the total volume to 16mL (80% of the total).
  - Stir the solution gently until the Tris base is completely dissolved.
- **Step 2:** Using a pH meter, measure the pH of each Tris solution and adjust the pH using either hydrochloric acid (HCl) to lower the pH or sodium hydroxide (NaOH) to raise the pH. **Record the volume of acid or base added to adjust the pH.**
- **Step 3:** Once you've hit your target pH, adjust the volume of the solution to 20mL with distilled water. Give it a good mix, then label your bottles clearly with the pH and Tris concentration.

#### Procedure: Preparation of Sodium Acetate/Acetic Acid Buffers

Let's do another example, but this time the buffer has two concentrations from different chemicals: Your goal is to create 0.1M sodium acetate/0.1M acetic acid solution at pH 5.5 and 7.5. Here's a step-by-step guide, or Standard Operating Procedure (SOP), for preparing a buffer with two separate concentrations:

- **Step 1:** Gather your materials and prepare a 10mL of 0.2M Sodium Acetate solution.
- **Step 2:** Gather your materials and prepare a 10mL of 0.2M Acetic acid solution.
- **Step 3:** Mix the two solutions together to obtain 20mL of volume.
- **Step 4:** Stir the solution gently and use a pH meter to adjust the pH using either HCl or NaOH until the desired pH of 5.5 or 7.5 is achieved. **Record the volume of acid or base added to adjust the pH.**
- **Step 5:** Once you've hit your target pH, adjust the volume of the solution to 20mL with distilled water. Give it a good mix, then label your bottles clearly with the pH and Tris concentration.

And there you have it! That's the basics of pH adjustment in the lab. Remember, practice makes perfect, so don't be afraid to get your hands dirty and experiment with different pH levels. Who knows, you might just discover the next big breakthrough in science!

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#### LAB ACTIVITY - PART 3 - Preparation of Stock and Gel

Finally, let's dive into gel preparation, a crucial step in the realm of electrophoresis. **Electrophoresis** helps us separate DNA, RNA, or proteins based on their size and charge. Today, we're focusing on agarose gels.

Imagine you're in the lab, ready to separate DNA fragments using agarose gel electrophoresis. You probably want to make sure there's a buffer that can support the high-voltage current. TBE Buffer ensures a stable pH and provides the necessary ions for the electric performance during electrophoresis. Here's a step-by-step guide, or Standard Operating Procedure (SOP), for preparing the agarose gel and the required buffer:

The objective is to prepare solutions of calcium chloride, 1X TBE buffer, and agarose gels in 1X TBE buffer.

**Materials:**

- Calcium Chloride (Carolina, 851840)
- Tris base (Flinn Scientific, #FB2393)
- Boric acid (Flinn Scientific, #B0081)
- EDTA (Gold Biotechnology, #E-210)
- Agarose powder
- Distilled water
- Serological pipette or Micropipette
- Beakers
- Magnetic stirrer or stirring rod
- pH meter
- Microwave or hot plate
- Gel electrophoresis chamber

**Procedure: Preparation of 1X TBE Buffer****\*Step 1: Preparation of 30mL of 1X TBE Buffer, pH 8**

- Weigh to obtain 0.1M Tris, 0.1M boric acid, and 0.002M EDTA.
  - \*Uh Oh! It's extremely difficult to accurately weigh out the 0.002M EDTA. Thus, it is recommended that we start with a higher concentration of TBE buffer, and then dilute it later.

**Step 1: Preparation of 30mL of 10X TBE Buffer, pH 8**

- Let's start again: Weigh to obtain 1M Tris, 1M boric acid, and 0.02M EDTA.
- Dilute the mixture to a final volume of 30mL with distilled water.
- Stir the solution thoroughly until all components are dissolved.
- Check the pH of the solution using a pH meter and adjust if necessary using concentrated HCl or NaOH solutions.
- Don't forget to label your bottle clearly as 10X TBE buffer, pH 8.

**Step 2: Dilute 10X to 1X TBE Buffer, pH 8**

- Dilute the solution 10-fold, by transferring 3mL into 27mL distilled water to obtain 1X TBE buffer.
  - Apply  $C_1V_1=C_2V_2$  to create 30mL of 1X TBE buffer, pH 8.

**Step 3: Preparation of 0.8% Agarose Gels**

- Measure out 30mL of the 1X TBE buffer prepared from the previous steps.
- Add 0.8% of agarose to the TBE buffer solution.
  - Measure 0.24g of agarose powder for each 30mL 1X TBE buffer into a clean container.
- Heat the mixture in a microwave (40 seconds interval) or hotplate until the agarose is completely dissolved, stirring occasionally.
- Let the solution cool to around 42°C before pouring it into a gel tray.
- Insert a comb into the gel to create wells – the special spots for sample loading.
  - Pour the solution into a gel tray and insert a comb to create wells for sample loading.
- Allow the gel to solidify at room temperature for roughly 30 minutes.

The agarose solution, now solidified, forms a gel matrix with wells, ready for molecular biology experiments. Once the gel is solidified, we place the gel into the electrophoresis chamber filled with 1X TBE buffer, creating a conducive environment. Then,

our DNA samples can be loaded into the wells for their separation.

#### Other: Preparation of 0.2M Calcium Chloride Solution

- Create a stock solution of Calcium Chloride
  - Combine 1.11g of calcium chloride with 10mL of distilled water in a clean beaker
  - Stir the solution until the calcium chloride is completely dissolved.
- Preparing 1mL of 0.1M Calcium Chloride:
  - Dilute and aliquot from the calcium chloride stock solution to obtain 1mL of 0.2M Calcium chloride solution.

## LAB ACTIVITY - PART 4: Practicum 2 (60 points total)

This evaluation will assess your proficiency in various laboratory techniques. You will be split into TWO teams. Work together to complete all projects efficiently and accurately. Work together within your team to plan out each task. Begin assigning tasks for every individual! Please follow the Standard Operating Procedures (SOP) from the previous class carefully. Your team will create one **of each buffer** from the list below. Ensure all submitted buffers are labeled, otherwise, the buffer will be 0 points

### Objective:

- **Team Buffer-Making Challenge:** This practicum tests your hands-on lab skills and your ability to work as a team under time pressure.
- You will work in teams to create 10 different buffers/solutions using only the provided SOPs—no pH meters allowed! Your solutions must be labeled clearly to receive credit.

#### Team Structure:

- You will be divided into **two teams**.
- Each team will complete **all 10 buffer preparations**.
- **Maximum: 6 students per team.**
- Everyone must be assigned specific tasks—communication and collaboration are key!

#### Time Breakdown:

- **First 30 minutes** → Discussion, equipment gathering, task planning.
- **Next 60 minutes** → You must complete the buffer preparations during this time.

| Order number (#) | Buffer and concentration             | pH  | Volume (mL) |
|------------------|--------------------------------------|-----|-------------|
| #1               | Luria-Bertani (LB) Broth             | N/A | 50mL        |
| #2               | Luria-Bertani (LB) Agar Plate        | N/A | 30mL        |
| #3               | 0.1M Calcium Chloride                | N/A | 1mL         |
| #4               | 0.8% Agarose Gels                    | N/A | 30mL        |
| #5               | 1X TBE buffers                       | 8   | 30mL        |
| #6               | 0.5M Tris                            | 6   | 20mL        |
| #7               | 0.5M Tris                            | 7   | 20mL        |
| #8               | 0.5M Tris                            | 8   | 20mL        |
| #9               | 0.1M Sodium Acetate/0.1M Acetic Acid | 7.5 | 20mL        |
| #10              | 0.1M Sodium Acetate/0.1M Acetic Acid | 5.5 | 20mL        |

### Part 1 – Volume (20 points)

- Create order 1 and 3.
- Follow the SOP for volume measurement precisely.
- Grade: Your team's performance will be graded based on the average percent error calculated from measured volume weights. (1% error = 19/20pts)

### Part 2 – Autoclave and Gel (10 points)

- Create orders 2 and 4.
- Follow the SOP for autoclaving and gel preparation provided.
- Grade: Solidified solutions will earn 5 points each, while non-solid will receive 0 points.

### Part 3 – pH Measurements (30 points)

- Create orders 5 through 10 without using any pH meter. The volume of acid or base should already been measured and noted from the lab activity part 2.
- Your team's performance will be graded based on the average percent error on pH measurements. (1% error = 29/30pts)

### General information:

- Each part of the evaluation is time-sensitive, so manage your time wisely.
- Maintain cleanliness and organization throughout the evaluation.
- Follow all safety protocols and wear appropriate personal protective equipment.
- Any deviation from the provided SOP may result in point deductions.
- Upon completion, submit your results to the instructor for evaluation.
- Good luck! Your performance in this evaluation will demonstrate your mastery of laboratory techniques and attention to detail.

#### Buffer List

##### Bacterial Transformation

- 50mL Luria-Bertani (LB) Broth
  - Add the following into 125mL Erlenmeyer Flask (Carolina, #721149A)
    - 1% of tryptone (Carolina, #216741)
    - 0.5% of yeast extract (Carolina, #216746)
    - 1% of NaCl (Flinn Scientific, #S0064)
  - Capped the flasks with aluminum foil (Carolina, #974844)
  - Autoclaved (25 minutes, 120oC)
- 30mL Luria-Bertani (LB) Agar Plate
  - Add the following into 125mL Erlenmeyer Flask (Carolina, #721149A)
    - 3% Luria Broth Agar Base (Carolina, 216701)
  - Capped the flasks with aluminum foil (Carolina, #974844)
  - Autoclaved (25 minutes, 120oC)
  - After cooling down to 42oC, pour it into a petri dish (Carolina, # 741251).
- 1mL of 0.1M Calcium Chloride (Carolina, 851840)

##### Restriction Digest and Electrophoresis

- 30mL of 0.8% Agarose Gels in 1X TBE buffer
- 30mL of 1X TBE buffers, pH 8
  - 0.1M Tris (Flinn Scientific, #FB2393)
  - 0.1M Boric acid (Flinn Scientific, #B0081)
  - 0.002M EDTA (Gold Biotechnology, #E-210)

##### Others



- 20mL 0.5M Tris, pH 4 (Flinn Scientific, #FB2393)
- 20mL 0.5M Tris, pH 7 (Flinn Scientific, #FB2393)
- 20mL 0.5M Tris, pH 10 (Flinn Scientific, #FB2393)
- 20mL of 0.1M Sodium Acetate/0.1 Acetic acid, pH 7.5
- 20mL of 0.1M Sodium Acetate/0.1 Acetic acid, pH 5.5

### ? Post-Lab Questions

1. Did the pH of the media change after autoclaving? If so, what could be the reason for this change?
2. How did the buffer solution respond to the addition of acids and bases? Explain the chemical basis for this behavior.
3. What challenges did you encounter during the pH adjustment process, and how did you address them?
4. Reflect on the importance of precise pH control in biological experiments and how it can affect experimental outcomes.
5. Summarize the key learnings from the lab, emphasizing the importance of sterilization, pH control, and buffer preparation in biological research.
6. Discuss how the skills and knowledge gained from this lab can be applied to future experiments

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## 13: Final Practicum

### Learning Objectives

- Understand the roles of consumers, producers, and QA/QC teams in buffer solution production and evaluation.
- Recognize the financial and quality control aspects of laboratory work.

#### Team Roles and Responsibilities:

- **Consumers:** Strategize purchasing decisions within an \$80 budget, ensuring cost-effective selection of buffer solutions.
- **Producers:** Accurately prepare buffer solutions while managing a \$20 budget for raw materials.
- **QA/QC and Supplier:** Supply raw materials to producers and verify the quality of produced buffer solutions.

#### Understand Financial Constraints:

- **Consumers:** Plan strategic purchasing to optimize buffer acquisition.
- **Producers:** Budget wisely for raw materials to minimize overspending penalties.

#### Quality Assurance Expectations:

- **Producers:** Ensure precision in buffer production to meet pH and volume requirements.
- **QA/QC:** Prepare to assess errors beyond 3% and document discrepancies.

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### LAB ACTIVITY - Final Practicum (80 points)

**Objective:** This simulation helps you understand the roles of consumers, producers, and quality assurance/control teams in making and evaluating buffer solutions. It's a hands-on way to learn financial concepts and quality control principles in labs. Embrace the challenge and work together for the best results. Remember, each dollar spent equals one point, encouraging smart spending. Overall, it's a practical way to apply financial and quality control ideas, collaborate, and learn about different lab roles.

#### Timeline:

- Week 12: Introduction, Planning, Troubleshooting, and Collaborating (pH meter allowed)
- Week 13: Production Phase 1 (pH meter not allowed)
- Week 14: Production Phase 2 (pH meter not allowed)
- Week 15: Production Phase 3 (pH meter not allowed)

#### Teams:

- Consumers
- Producers
- QA/QC and Supplier

#### Consumer Instructions:

As consumers, your role is to make informed decisions when purchasing buffer solutions. This exercise emphasizes the importance of budget management and strategic decision-making, mirroring real-world laboratory practices. You will start with a hypothetical budget of \$80. Review the 10 buffer solutions listed below, carefully considering both cost and quantity and decide which buffers to purchase.

#### Grading Policy:

- **Unused Funds:** Any unused funds will be deducted from your team's overall points. For example, if your team spends \$75, your grade will be 75/80. This encourages efficient spending and highlights the importance of optimizing resources.
- **Refunds for Producer Errors:** If the **producer team** fails to fulfill an order or delivers a product with an error rate above 3%, they will lose 5 points per order. Those points will be transferred to your team to account for the inconvenience and emphasize the consequences of inadequate planning.

By stepping into the role of a consumer, you will gain a deeper understanding of the importance of resource allocation, procurement accuracy, and quality assurance in laboratory management.

| Buffer and Concentration             | pH  | Volume (mL) | Cost for one buffer (\$) | Cost for two or more buffers (\$) |
|--------------------------------------|-----|-------------|--------------------------|-----------------------------------|
| Luria-Bertani (LB) Broth             | N/A | 50mL        | \$5.00                   | \$2.50                            |
| Luria-Bertani (LB) Agar Plate        | N/A | 30mL        | \$5.00                   | \$2.50                            |
| 0.1M Calcium Chloride                | N/A | 1mL         | \$5.00                   | \$2.50                            |
| 0.8% Agarose Gels                    | N/A | 30mL        | \$5.00                   | \$2.50                            |
| 1X TBE Buffer                        | 8   | 30mL        | \$5.00                   | \$2.50                            |
| 0.5M Tris                            | 6   | 20mL        | \$5.00                   | \$2.50                            |
| 0.5M Tris                            | 7   | 20mL        | \$5.00                   | \$2.50                            |
| 0.5M Tris                            | 8   | 20mL        | \$5.00                   | \$2.50                            |
| 0.1M Sodium Acetate/0.1M Acetic acid | 7.5 | 20mL        | \$5.00                   | \$2.50                            |
| 0.1M Sodium Acetate/0.1M Acetic acid | 6.6 | 20mL        | \$5.00                   | \$2.50                            |

#### Producer Instructions:

As producers, your task is to create buffer solutions based on consumer orders while maintaining high quality and adhering to a strict budget. This activity emphasizes the importance of precision, efficiency, and customer satisfaction in production processes.

#### Your Role:

- Optional:** You may recalibrate the pH meter and scale prior to starting the production
- Review Consumer Orders:** Understand the buffer solution requirements provided by consumers and plan accordingly.
  - Example Consumer Request:
    - Tryptone Salt Broth Buffer Solution**
- Purchase Raw Materials:** Use your **\$20 budget** to buy the raw materials needed to fulfill the consumer orders.
  - Example Raw Material Purchase:
    - 0.5 g of Tryptone**
    - 2 g of NaCl**
  - Note: Purchasing slightly more raw materials than required is recommended to account for potential errors during production.*
- Produce High-Quality Buffer Solutions:** Once raw materials are secured, focus on creating accurate and reliable buffer solutions to meet consumer demands. Maintaining quality ensures your solutions are suitable for scientific use.
- Complete Orders During Class Time:** Ensure all production is completed within the allocated time to emphasize the importance of meeting deadlines in manufacturing.

#### Grading Policy:

- Overspending Penalty:** Spending beyond the **\$20 budget** will result in point deductions. For example, if your team spends **\$21**, you will lose **1 point**. This reinforces the need for financial discipline and resource efficiency.

- **Quality Penalty:** For each unfulfilled order or any buffer solution with an error rate above 3%, your team will lose 5 points. These points will be transferred to the consumer team's score to simulate the impact of production errors on customer satisfaction.

By stepping into the role of producers, you'll gain valuable insights into quality control, resource management, and the challenges of meeting customer expectations under real-world constraints.

| Raw Reagent and Materials | Costs per gram or mL (\$) |
|---------------------------|---------------------------|
| Tryptone                  | \$1.00                    |
| Yeast Extract             | \$0.50                    |
| Sodium Chloride           | \$1.00                    |
| Luria Broth Agar Base     | \$5.00                    |
| Calcium Chloride          | \$20.00                   |
| Agarose                   | \$1.00                    |
| Tris                      | \$1.00                    |
| Boric Acid                | \$1.00                    |
| EDTA                      | \$5.00                    |
| 0.5M Sodium Acetate       | \$0.50                    |
| 0.5M Acetic Acid          | \$0.50                    |

#### QA/QC and Supplier Instructions:

As QA/QC and Supplier, your role is two-fold: ensuring quality standards for buffer solutions and supplying raw materials to producers. This dual responsibility provides valuable insights into the importance of reliable supply chains and quality assurance in scientific research.

#### Task #1: Supply Raw Materials and Calculate Costs

1. **Provide Raw Materials to Producers (Team #2):** Producers will submit a supply request specifying the quantity of raw materials they need for creating buffer solutions.
  - Example Supply Request:
    - 0.5 g of Tryptone (\$0.50)
    - 2 g of NaCl (\$2.00)
    - Total Cost: \$2.50
2. **Record Transactions:** Document the requested materials and ensure the correct costs are calculated.

#### Task #2: Evaluate Buffer Quality

1. **Check Buffer Solutions from Producers:** Review each buffer solution for quality, ensuring that it meets the required standards.
2. **Record Errors:** Document any errors above 3%. If buffer solutions exceed a 3% **error rate**, producers will lose 5 points, reflecting the importance of quality control.

#### Rotation:

Teams will rotate roles weekly to gain diverse experiences. This rotation fosters a comprehensive understanding of laboratory processes, strengthens collaboration, and enhances problem-solving skills.

|                          |    |             |                  |
|--------------------------|----|-------------|------------------|
| Buffer and Concentration | pH | Volume (mL) | Grading Criteria |
|--------------------------|----|-------------|------------------|

|                                      |     |      |                       |
|--------------------------------------|-----|------|-----------------------|
| Luria-Bertani (LB) Broth             | N/A | 50mL | % error on weight (g) |
| Luria-Bertani (LB) Agar Plate        | N/A | 30mL | Solidification or not |
| 0.1M Calcium Chloride                | N/A | 1mL  | % error on weight     |
| 0.8% Agarose Gels                    | N/A | 30mL | Solidification or not |
| 1X TBE Buffer                        | 8   | 30mL | % error on pH         |
| 0.5M Tris                            | 6   | 20mL | % error on pH         |
| 0.5M Tris                            | 7   | 20mL | % error on pH         |
| 0.5M Tris                            | 8   | 20mL | % error on pH         |
| 0.1M Sodium Acetate/0.1M Acetic acid | 7.5 | 20mL | % error on pH         |
| 0.1M Sodium Acetate/0.1M Acetic acid | 6.6 | 20mL | % error on pH         |

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### ? Discussion Questions

1. How did financial constraints influence decision-making in each role?
2. What were the primary sources of error in buffer preparation, and how could they be minimized?
3. How did quality control impact consumer confidence and overall team performance?
4. What strategies could improve teamwork and efficiency in future lab settings?
5. Summarize key takeaways from the simulation regarding resource management and quality control.
6. Identify areas for improvement and propose strategies for optimizing performance in future laboratory environments.
7. Reflect on how this exercise relates to real-world laboratory operations and financial decision-making.

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## 14: Materials List

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### Week 1 – Lab Safety (Will be required throughout the semester)

- Gloves (XS, S, M, L, and XL)
- Disinfectants
- Lab coat

### Week 2 – Basic Measuring Techniques

- Measuring scale or analytical scale
- Weigh boats (Carolina, 702332)
- Weigh papers (Thomas Scientific, 1229D27)
- Scoopula (Carolina, 702851)
- 50mL beaker (Carolina, 717921)
- 250mL graduate cylinders (Carolina, 721604)
- 25mL Serological pipette (USA Scientific, 1072-5410)
- 25mL Pipet pump (Carolina, 736864)
- Standard mass
- NaCl (Flinn Scientific, S0064)

### Week 3 – Micro pipetting and Calibration.

- Same as Week 2.
- Micropipettes sets.
- Micropipette tips
- 1.5mL-microcentrifuge tube (Carolina, #215244)
- 50mL-High Speed conical tubes (Fisher Scientific, #352070)
- 15mL-centrifuge tube (Carolina, # 215085)
- 50mL Conical centrifuge tubes (Fisher Scientific, 352070)
- Four colored waters (Carolina, # 898030)
  - Have water stock in 500mL beaker (Carolina, 717924)

### Week 4 – Dilutions and Spectrophotometer

- Same as Week 3
- 70% Ethanol spray (Flinn Scientific, E0013)
- Spectrophotometry
- Cuvette holders (Grainger, 12C181)
- Uvettes (Fisher Scientific, E4099100008)
- 500mL of 0.5M of Copper (II) Sulfate (Flinn Scientific, C0108)
- 1M Sodium acetate (Flinn Scientific, #S0038)
- 1M Acetic Acid (Flinn Scientific, #A0187)

### Week 5 – pH Calibration and buffer measurements

- Same as Week 3.
- Magnetic Stir (Carolina, #701091)
- Hot Plate Stirrer Machine (Carolina, #701012)
- Stir Bar retriever (Carolina, #701089)
- pH Meter (Carolina, #186032)
- 6M Hydrochloric Acids (HCl) (Flinn Scientific, #H0033)
- 6M Sodium Hydroxide (NaOH) (Flinn Scientific, #S0242)
- pH paper (Carolina, 894720)
- pH Calibration buffers (**Make sure they are NEW**)
- pH Electrode storage solution (Carolina, # 185763)
- pH Electrode Cleaning solution (Carolina, # 185764)

## Week 6 to Week 15 – Advance Buffer Preparations

- Same as Week 2.
- Same as Week 3.
- 125mL Erlenmeyer Flask (Carolina, #721149A)
- Tryptone (Carolina, #216741)
- Yeast extract (Carolina, #216746)
- NaCl (Flinn Scientific, #S0064)
- Aluminum foil (Carolina, #974844)
- Luria Broth Agar Base (Carolina, 216701)
- Petri dish (Carolina, # 741251)
- Calcium Chloride (Carolina, 851840)
- Tris (Flinn Scientific, #FB2393)
- Boric acid (Flinn Scientific, #B0081)
- EDTA (Gold Biotechnology, #E-210)
- Agarose
- MiniONE agarose gel electrophoresis kit

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Proper handling and disposal of hazardous materials and chemicals are essential to ensure laboratory safety and environmental protection. Below are guidelines for the reagents and supplies listed in your lab activities:

### General Safety Measures:

1. **Personal Protective Equipment (PPE):** Always wear appropriate PPE, including gloves, lab coats, and safety goggles, when handling chemicals.
2. **Ventilation:** Work in well-ventilated areas, preferably under a fume hood, to minimize inhalation risks.
3. **Training:** Ensure all personnel are trained in chemical safety and emergency procedures.

### Handling and Disposal Guidelines:

1. **Gloves (XS, S, M, L, XL):**
  - **Handling:** Use gloves suitable for the chemicals being handled.
  - **Disposal:** Dispose of used gloves in designated hazardous waste containers.
2. **Disinfectants:**
  - **Handling:** Follow manufacturer instructions for dilution and application.
  - **Disposal:** Dispose of unused disinfectants according to institutional guidelines, typically as hazardous waste.
3. **Lab Coats:**
  - **Handling:** Wear lab coats to protect against chemical spills.
  - **Disposal:** If contaminated with hazardous materials, launder separately or dispose of as hazardous waste.
4. **Measuring Scale or Analytical Scale:**
  - **Handling:** Calibrate regularly and clean after each use.
5. **Weigh Boats and Weigh Papers:**
  - **Handling:** Use for weighing chemicals to prevent contamination.
  - **Disposal:** Dispose of contaminated weigh boats and papers in hazardous waste containers.
6. **Scoopula:**
  - **Handling:** Use for transferring solid chemicals.
  - **Disposal:** Clean thoroughly after use
7. **Beakers and Graduated Cylinders:**
  - **Handling:** Use for mixing and measuring liquids.
  - **Disposal:** Clean with appropriate solvents; dispose of cleaning solutions as hazardous waste if they contain hazardous residues.
8. **Serological Pipettes and Pipet Pumps:**

- **Handling:** Use for transferring liquids; ensure proper technique to avoid contamination.
- **Disposal:** Dispose of used pipette tips in designated sharps containers; clean and reuse pipet pumps.

9. **Standard Mass:**

- **Handling:** Use for calibration; handle with care to maintain accuracy.
- **Disposal:** If damaged or contaminated, consult with a supervisor.

10. **Sodium Chloride (NaCl):**

- **Handling:** Use as directed; avoid inhalation of dust.
- **Disposal:** Dispose of excess NaCl in regular waste unless contaminated.

11. **Micropipettes and Tips:**

- **Handling:** Use for precise liquid measurements; avoid contamination.
- **Disposal:** Dispose of contaminated tips in biohazard waste containers; clean and calibrate micropipettes regularly.

12. **Centrifuge Tubes (1.5mL, 15mL, 50mL):**

- **Handling:** Use for sample separation; ensure proper sealing.
- **Disposal:** Dispose of used tubes in biohazard waste containers if contaminated.

13. **Colored Waters:**

- **Handling:** Use as tracers or standards; handle according to chemical properties.
- **Disposal:** Dispose of in the sink.

14. **70% Ethanol Spray:**

- **Handling:** Use for surface disinfection; avoid inhalation and open flames.
- **Disposal:** Dispose of unused ethanol in designated hazardous waste containers.

15. **Spectrophotometry Equipment:**

- **Handling:** Use for measuring light absorption; handle with care to avoid damage.
- **Disposal:** Follow manufacturer's guidelines for maintenance.

16. **Cuvette Holders and Uvettes:**

- **Handling:** Use for holding samples in spectrophotometry; clean after each use.
- **Disposal:** Dispose of broken or contaminated cuvettes in waste containers.

17. **Copper (II) Sulfate (0.5M):**

- **Handling:** Use with caution; avoid inhalation and skin contact.
- **Disposal:** Dispose of excess solution in designated hazardous waste containers.

18. **Sodium Acetate (1M) and Acetic Acid (1M):**

- **Handling:** Use as buffers; handle with care to avoid spills.
- **Disposal:** Dispose of according to institutional guidelines; typically as hazardous waste.

19. **Magnetic Stirrer and Hot Plate Stirrer Machine:**

- **Handling:** Always ensure that these devices are turned off when not in use. Handle carefully to avoid breakage or electrical hazards.
- **Disposal:** Follow manufacturer's guidelines for maintenance.

20. **Stir Bar Retriever:**

- **Handling:** Use for retrieving stir bars from containers. Handle carefully to avoid damaging the stir bars.
- **Disposal:** If stir bars are damaged, consult with a supervisor.

21. **pH Meter:**

- **Handling:** Calibrate and store according to manufacturer's instructions to ensure accurate measurements. Avoid contact with the electrode unless necessary.
- **Disposal:** If damaged, consult with a supervisor. Never dispose of pH electrodes in general waste.

22. **125mL Erlenmeyer Flask:**

- **Handling:** Use for mixing or heating liquids. Avoid exposure to extreme heat or cold.
- **Disposal:** Clean thoroughly after use.

23. **Tryptone, Yeast Extract, NaCl, Luria Broth Agar Base:**



- **Handling:** These reagents are generally non-hazardous but should be handled carefully to avoid contamination.
- **Disposal:** Dispose of excess chemicals in designated chemical waste containers.

24. **Aluminum Foil:**

- **Handling:** Handle with clean hands or gloves to avoid contamination.
- **Disposal:** Dispose of clean foil in regular waste. If contaminated, dispose of it in a designated hazardous waste container.

25. **Petri Dish:**

- **Handling:** Use for agar plating. Sterilize before and after use.
- **Disposal:** Dispose of used or broken petri dishes in designated biohazard waste containers if contaminated.

26. **Calcium Chloride (CaCl<sub>2</sub>), Tris, Boric Acid, EDTA:**

- **Handling:** Handle with care and wear appropriate PPE. Store in tightly sealed containers.
- **Disposal:** Dispose of excess chemicals according to institution-specific waste disposal protocols.

27. **Agarose:**

- **Handling:** Handle with care and avoid inhalation of the powder.
- **Disposal:** Dispose of agarose in general waste unless contaminated with hazardous chemicals.

28. **MiniONE Agarose Gel Electrophoresis Kit:**

- **Handling:** Follow the manufacturer's instructions for setting up and running electrophoresis. Always handle with gloves to avoid contamination.
- **Disposal:** Dispose of used gels in biohazard waste containers. Ensure all electrophoresis chemicals are disposed of in chemical waste containers.

29. **6M Hydrochloric Acid (HCl):**

- **Handling:** Wear gloves, goggles, and lab coat. Always add acid to water (not the other way around) to prevent exothermic reactions.
- **Disposal:** Neutralize with a base like sodium bicarbonate before disposal down the drain (ensure this is approved by local regulations).

30. **6M Sodium Hydroxide (NaOH):**

- **Handling:** Handle with care, as NaOH is a strong base and can cause burns. Always use in a well-ventilated area.
- **Disposal:** Neutralize with an acid like hydrochloric acid before disposal in compliance with institutional guidelines.

31. **pH Paper:**

- **Handling:** Store in a dry place. Handle with clean hands to avoid contamination.
- **Disposal:** Dispose of used pH paper in general waste.

32. **pH Calibration Buffers:**

- **Handling:** Use freshly prepared or new calibration buffers for accuracy.
- **Disposal:** Dispose of used calibration buffers in appropriate chemical waste containers.

33. **pH Electrode Storage Solution and Cleaning Solution:**

- **Handling:** Use only as directed to prevent damage to the electrode.
- **Disposal:** Dispose of any excess solution according to institutional hazardous waste procedures.

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